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(54) **DISPLAY ASSEMBLIES WITH
CONDENSATION MITIGATION AND
RELATED SYSTEMS AND METHODS**

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(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,093,355 A 6/1978 Kaplit et al.
4,593,978 A 6/1986 Mourey et al.
(Continued)

FOREIGN PATENT DOCUMENTS

AU 2010218083 B2 10/2016
AU 2016203550 B2 3/2018
(Continued)

OTHER PUBLICATIONS

University of Miami, Calculate Temperature, Dewpoint, or Relative
Humidity, <http://bmcnoldy.rsmas.miami.edu/Humidity.html>, URL
accessed Apr. 18, 2022, 1 page.

(Continued)

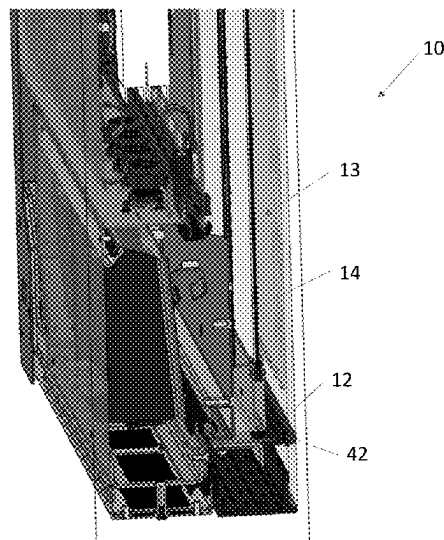
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(57) **ABSTRACT**

Display assemblies and related systems and methods are
provided which provide condensation mitigation. A cover
layer forms a forward portion of a housing for an electronic
display located rearward of the cover layer. A continuous
airflow pathway extends entirely within the housing such
that the continuous airflow pathway is partitioned from an
ambient environment. One or more gaskets are provided
which, at least in part, partition the continuous airflow
pathway from the ambient environment. The one or more
gaskets extending from an interior surface of the cover layer
to a portion of the housing.

20 Claims, 14 Drawing Sheets



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(56) **References Cited**

U.S. PATENT DOCUMENTS

4,634,225	A	1/1987	Haim et al.	7,236,154	B1	6/2007	Kerr et al.
4,722,669	A	2/1988	Kundert	7,307,614	B2	12/2007	Vinn
5,029,982	A	7/1991	Nash	7,324,080	B1	1/2008	Hu et al.
5,086,314	A	2/1992	Aoki et al.	7,330,002	B2	2/2008	Joung
5,088,806	A	2/1992	McCartney et al.	7,354,159	B2	4/2008	Nakamura et al.
5,162,785	A	11/1992	Fagard	7,447,018	B2	11/2008	Lee et al.
5,247,374	A	9/1993	Terada	7,474,294	B2	1/2009	Leo et al.
5,285,677	A	2/1994	Oehler	7,480,042	B1	1/2009	Phillips et al.
5,559,614	A	9/1996	Urbish et al.	7,518,600	B2	4/2009	Lee
5,661,374	A	8/1997	Cassidy et al.	7,595,785	B2	9/2009	Jang
5,748,269	A	5/1998	Harris et al.	7,639,220	B2	12/2009	Yoshida et al.
5,767,489	A	6/1998	Ferrier	7,659,676	B2	2/2010	Hwang
5,769,705	A	6/1998	O'Callaghan et al.	7,692,621	B2	4/2010	Song
5,783,909	A	7/1998	Hochstein	7,724,247	B2	5/2010	Yamazaki et al.
5,786,801	A	7/1998	Ichise	7,795,574	B2	9/2010	Kennedy et al.
5,808,418	A	9/1998	Pitman et al.	7,795,821	B2	9/2010	Jun
5,818,010	A	10/1998	McCann	7,800,706	B2	9/2010	Kim et al.
5,952,992	A	9/1999	Helms	7,804,477	B2	9/2010	Sawada et al.
5,962,933	A	10/1999	Henderson et al.	7,982,706	B2	7/2011	Ichikawa et al.
5,991,153	A *	11/1999	Heady	8,087,787	B2	1/2012	Medin
			H05K 7/202	8,111,371	B2	2/2012	Suminoe et al.
			361/679.48	8,125,163	B2	2/2012	Dunn et al.
6,085,152	A	7/2000	Doerfel	8,144,110	B2	3/2012	Huang
6,089,751	A	7/2000	Conover et al.	8,175,841	B2	5/2012	Ooghe
6,144,359	A	11/2000	Grave	8,194,031	B2	6/2012	Yao et al.
6,153,985	A	11/2000	Grossman	8,248,203	B2	8/2012	Hanwright et al.
6,157,143	A	12/2000	Bigio et al.	8,319,936	B2	11/2012	Yoshida et al.
6,157,432	A	12/2000	Helbing	8,325,057	B2	12/2012	Salter
6,181,070	B1	1/2001	Dunn et al.	8,352,758	B2	1/2013	Atkins et al.
6,191,839	B1	2/2001	Briley et al.	8,508,155	B2	8/2013	Schuch
6,259,492	B1	7/2001	Imoto et al.	8,569,910	B2	10/2013	Dunn et al.
6,292,228	B1	9/2001	Cho	8,605,121	B2	12/2013	Chu et al.
6,297,859	B1	10/2001	George	8,643,589	B2	2/2014	Wang
6,380,853	B1	4/2002	Long et al.	8,700,226	B2	4/2014	Schuch et al.
6,388,388	B1	5/2002	Weindorf et al.	8,797,372	B2	8/2014	Liu
6,400,101	B1	6/2002	Biebl et al.	8,810,501	B2	8/2014	Budzelaar et al.
6,417,900	B1	7/2002	Shin et al.	8,823,630	B2	9/2014	Roberts et al.
6,496,236	B1	12/2002	Cole et al.	8,829,815	B2	9/2014	Dunn et al.
6,509,911	B1	1/2003	Shimotono	8,895,836	B2	11/2014	Amin et al.
6,535,266	B1	3/2003	Nemeth et al.	8,901,825	B2	12/2014	Reed
6,556,258	B1	4/2003	Yoshida et al.	8,982,013	B2	3/2015	Sako et al.
6,628,355	B1	9/2003	Takahara	8,983,385	B2	3/2015	Macholz
6,701,143	B1	3/2004	Dukach et al.	8,988,011	B2	3/2015	Dunn
6,712,046	B2	3/2004	Nakamichi	9,030,129	B2	5/2015	Dunn et al.
6,753,661	B2	6/2004	Muthu et al.	9,167,655	B2	10/2015	Dunn et al.
6,753,842	B1	6/2004	Williams et al.	9,286,020	B2	3/2016	Dunn et al.
6,762,741	B2	7/2004	Weindorf	9,400,192	B1	7/2016	Salser, Jr. et al.
6,798,341	B1	9/2004	Eckel et al.	9,445,470	B2	9/2016	Wang et al.
6,809,718	B2	10/2004	Wei et al.	9,448,569	B2	9/2016	Schuch et al.
6,812,851	B1	11/2004	Dukach et al.	9,451,060	B1	9/2016	Bowers et al.
6,813,375	B2	11/2004	Armato, III et al.	9,516,485	B1	12/2016	Bowers et al.
6,839,104	B2	1/2005	Taniguchi et al.	9,536,325	B2	1/2017	Bray et al.
6,850,209	B2	2/2005	Mankins et al.	9,622,392	B1	4/2017	Bowers et al.
6,885,412	B2	4/2005	Ohnishi et al.	9,799,306	B2	10/2017	Dunn et al.
6,886,942	B2	5/2005	Okada et al.	9,867,253	B2	1/2018	Dunn et al.
6,891,135	B2	5/2005	Pala et al.	9,881,528	B2	1/2018	Dunn
6,943,768	B2	9/2005	Cavanaugh et al.	9,924,583	B2	3/2018	Schuch et al.
6,982,686	B2	1/2006	Miyachi et al.	10,194,562	B2	1/2019	Shelnutt et al.
6,996,460	B1	2/2006	Krahnstoeve et al.	10,255,884	B2	4/2019	Dunn et al.
7,015,470	B2	3/2006	Faytlin et al.	10,321,549	B2	6/2019	Schuch et al.
7,038,186	B2	5/2006	De Brabander et al.	10,409,544	B2	9/2019	Park et al.
7,064,733	B2	6/2006	Cok et al.	10,412,816	B2	9/2019	Schuch et al.
7,083,285	B2	8/2006	Hsu et al.	10,440,790	B2	10/2019	Dunn et al.
7,136,076	B2	11/2006	Evanicky et al.	10,578,658	B2	3/2020	Dunn et al.
7,174,029	B2	2/2007	Agostinelli et al.	10,586,508	B2	3/2020	Dunn
7,176,640	B2	2/2007	Tagawa	10,593,255	B2	3/2020	Schuch et al.
				10,607,520	B2	3/2020	Schuch et al.
				10,782,276	B2	9/2020	Dunn et al.
				10,795,413	B1	10/2020	Dunn
				10,803,783	B2	10/2020	Wang et al.
				10,858,886	B2	12/2020	Fasi et al.
				10,860,141	B2	12/2020	Wang et al.
				11,016,547	B2	5/2021	Whitehead et al.
				11,022,635	B2	6/2021	Dunn et al.
				11,132,715	B2	9/2021	Menendez et al.
				11,293,908	B2	4/2022	Dunn et al.
				11,526,044	B2	12/2022	Dunn
				11,656,255	B2	5/2023	Dunn et al.
				11,774,428	B2	10/2023	Dunn et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

11,815,755	B2	11/2023	Dunn	2008/0019147	A1	1/2008	Erchak et al.
11,977,065	B2	5/2024	Dunn et al.	2008/0055297	A1	3/2008	Park
12,007,637	B2	6/2024	Dunn	2008/0074382	A1	3/2008	Lee et al.
12,022,635	B2	6/2024	Dunn et al.	2008/0078921	A1	4/2008	Yang et al.
12,027,132	B1	7/2024	Dunn et al.	2008/0084166	A1	4/2008	Tsai
2002/0009978	A1	1/2002	Dukach et al.	2008/0111958	A1	5/2008	Kleverman et al.
2002/0020090	A1	2/2002	Sanders	2008/0136770	A1	6/2008	Peker et al.
2002/0050974	A1	5/2002	Rai et al.	2008/0143187	A1	6/2008	Hoekstra et al.
2002/0065046	A1	5/2002	Mankins et al.	2008/0151082	A1	6/2008	Chan
2002/0084891	A1	7/2002	Mankins et al.	2008/0165203	A1	7/2008	Pantfoerder
2002/0101553	A1	8/2002	Enomoto et al.	2008/0170031	A1	7/2008	Kuo
2002/0112026	A1	8/2002	Fridman et al.	2008/0176345	A1	7/2008	Yu et al.
2002/0126248	A1	9/2002	Yoshida	2008/0185976	A1	8/2008	Dickey et al.
2002/0154138	A1	10/2002	Wada et al.	2008/0204375	A1	8/2008	Shin et al.
2002/0164962	A1	11/2002	Mankins et al.	2008/0218501	A1	9/2008	Diamond
2002/0167637	A1	11/2002	Burke et al.	2008/0224892	A1	9/2008	Bogolea et al.
2002/0190972	A1	12/2002	Ven de Van	2008/0230497	A1	9/2008	Strickland et al.
2003/0007109	A1	1/2003	Park	2008/0246871	A1	10/2008	Kupper et al.
2003/0088832	A1	5/2003	Agostinelli et al.	2008/0259198	A1	10/2008	Chen et al.
2003/0122810	A1	7/2003	Tsirkel et al.	2008/0266554	A1	10/2008	Sekine et al.
2003/0204342	A1	10/2003	Law et al.	2008/0278099	A1	11/2008	Bergfors et al.
2003/0214242	A1	11/2003	Berg-johansen	2008/0278100	A1	11/2008	Hwang
2003/0230991	A1	12/2003	Muthu et al.	2008/0303918	A1	12/2008	Keithley
2004/0032382	A1	2/2004	Cok et al.	2009/0009997	A1	1/2009	Sanfilippo et al.
2004/0036622	A1	2/2004	Dukach et al.	2009/0014548	A1	1/2009	Criss et al.
2004/0036697	A1	2/2004	Kim et al.	2009/0033612	A1	2/2009	Roberts et al.
2004/0036834	A1	2/2004	Ohnishi et al.	2009/0079416	A1	3/2009	Vinden et al.
2004/0113044	A1	6/2004	Ishiguchi	2009/0085859	A1	4/2009	Song
2004/0165139	A1	8/2004	Anderson et al.	2009/0091634	A1	4/2009	Kennedy et al.
2004/0201547	A1	10/2004	Takayama	2009/0104989	A1	4/2009	Williams et al.
2004/0243940	A1	12/2004	Lee et al.	2009/0109129	A1	4/2009	Cheong et al.
2005/0012734	A1	1/2005	Johnson et al.	2009/0135167	A1	5/2009	Sakai et al.
2005/0024538	A1	2/2005	Park et al.	2009/0152445	A1	6/2009	Gardner, Jr.
2005/0043907	A1	2/2005	Eckel et al.	2009/0278766	A1	11/2009	Sako et al.
2005/0049729	A1	3/2005	Culbert et al.	2009/0284457	A1	11/2009	Botzas et al.
2005/0073518	A1	4/2005	Bontempi	2009/0289968	A1	11/2009	Yoshida
2005/0094391	A1	5/2005	Campbell et al.	2010/0033413	A1	2/2010	Song et al.
2005/0127796	A1	6/2005	Olesen et al.	2010/0039366	A1	2/2010	Hardy
2005/0140640	A1	6/2005	Oh et al.	2010/0039414	A1	2/2010	Bell
2005/0184983	A1	8/2005	Brabander et al.	2010/0039440	A1	2/2010	Tanaka et al.
2005/0231457	A1	10/2005	Yamamoto et al.	2010/0060861	A1	3/2010	Medin
2005/0242741	A1	11/2005	Shiota et al.	2010/0066484	A1	3/2010	Hanwright et al.
2006/0007107	A1	1/2006	Ferguson	2010/0177750	A1	7/2010	Essinger et al.
2006/0022616	A1	2/2006	Furukawa et al.	2010/0194725	A1	8/2010	Yoshida et al.
2006/0038511	A1	2/2006	Tagawa	2010/0231602	A1	9/2010	Huang
2006/0049533	A1	3/2006	Kamoshita	2010/0237697	A1	9/2010	Dunn et al.
2006/0087521	A1	4/2006	Chu et al.	2010/0253660	A1	10/2010	Hashimoto
2006/0125773	A1	6/2006	Ichikawa et al.	2010/0309361	A1	12/2010	Fukushima
2006/0130501	A1	6/2006	Singh et al.	2011/0001898	A1*	1/2011	Mikubo H05K 7/20972 349/58
2006/0197474	A1	9/2006	Olsen	2011/0016006	A1	1/2011	Opdycke
2006/0197735	A1	9/2006	Vuong et al.	2011/0032285	A1	2/2011	Yao et al.
2006/0207730	A1	9/2006	Berman et al.	2011/0032489	A1	2/2011	Kimoto et al.
2006/0214904	A1	9/2006	Kimura et al.	2011/0050738	A1	3/2011	Fujioka et al.
2006/0215044	A1	9/2006	Masuda et al.	2011/0058326	A1	3/2011	Idems et al.
2006/0220571	A1	10/2006	Howell et al.	2011/0074737	A1	3/2011	Hsieh et al.
2006/0238531	A1	10/2006	Wang	2011/0074803	A1	3/2011	Kerofsky
2006/0244702	A1	11/2006	Yamazaki et al.	2011/0102630	A1	5/2011	Rukes
2007/0013828	A1	1/2007	Cho et al.	2011/0148904	A1	6/2011	Kotani
2007/0047808	A1	3/2007	Choe et al.	2011/0163691	A1	7/2011	Dunn
2007/0152949	A1	7/2007	Sakai	2011/0175872	A1	7/2011	Chuang et al.
2007/0153117	A1	7/2007	Lin et al.	2011/0193872	A1	8/2011	Biernath et al.
2007/0171647	A1	7/2007	Artwohl et al.	2011/0231676	A1	9/2011	Atkins et al.
2007/0173297	A1	7/2007	Cho et al.	2011/0260534	A1	10/2011	Rozman et al.
2007/0200513	A1	8/2007	Ha et al.	2011/0264273	A1	10/2011	Grabinger et al.
2007/0222730	A1	9/2007	Kao et al.	2011/0279426	A1	11/2011	Imamura et al.
2007/0230167	A1	10/2007	McMahon et al.	2011/0283199	A1	11/2011	Schuch et al.
2007/0242153	A1	10/2007	Tang et al.	2012/0075362	A1	3/2012	Ichioaka et al.
2007/0247594	A1	10/2007	Tanaka	2012/0081279	A1	4/2012	Greenebaum et al.
2007/0268234	A1	11/2007	Wakabayashi et al.	2012/0176420	A1	7/2012	Liu
2007/0268241	A1	11/2007	Nitta et al.	2012/0182278	A1	7/2012	Ballestad
2007/0273624	A1	11/2007	Geelen	2012/0197459	A1	8/2012	Fukano
2007/0279369	A1	12/2007	Yao et al.	2012/0211001	A1	8/2012	Elshafei
2007/0291198	A1	12/2007	Shen	2012/0212520	A1	8/2012	Matsui et al.
2007/0297163	A1	12/2007	Kim et al.	2012/0252495	A1	10/2012	Moeglein et al.
2007/0297172	A1	12/2007	Furukawa et al.	2012/0268436	A1	10/2012	Chang
				2012/0269382	A1	10/2012	Kiyohara et al.
				2012/0284547	A1	11/2012	Culbert et al.
				2013/0027370	A1	1/2013	Dunn et al.

(56)	References Cited			2023/0333423 A1	10/2023	Dunn et al.
	U.S. PATENT DOCUMENTS			2023/0384277 A1	11/2023	Dunn et al.
				2024/0036379 A1	2/2024	Dunn
				2024/0106343 A1	3/2024	Nozawa
2013/0070567 A1	3/2013	Marzouq		FOREIGN PATENT DOCUMENTS		
2013/0098425 A1	4/2013	Amin et al.		AU	2016262614 B2	1/2019
2013/0107143 A1 *	5/2013	Dighde	G02F 1/133308	AU	2016308187 B2	2/2020
			349/1	CA	2754371 C	11/2017
2013/0113973 A1	5/2013	Miao		CA	2849902 C	2/2019
2013/0158730 A1	6/2013	Yasuda et al.		CA	2985673 C	3/2021
2013/0278868 A1	10/2013	Dunn et al.		CN	112526806 A	3/2021
2013/0279090 A1	10/2013	Brandt		EP	0313331 B1	2/1994
2013/0344794 A1	12/2013	Shaw et al.		EP	1686777 A1	8/2006
2014/0002747 A1	1/2014	Macholz		EP	2299723 A1	3/2011
2014/0132796 A1	5/2014	Prentice et al.		EP	2401738 A2	1/2012
2014/0139116 A1	5/2014	Reed		EP	2769376 A1	8/2014
2014/0184980 A1	7/2014	Onoue		EP	2577389 B1	5/2017
2014/0190240 A1	7/2014	He et al.		EP	3295452 A1	3/2018
2014/0204452 A1	7/2014	Branson		EP	2401738 B1	5/2018
2014/0232709 A1	8/2014	Dunn et al.		EP	3338273 A1	6/2018
2014/0293605 A1	10/2014	Chemel et al.		EP	4309347 A1	1/2024
2014/0365965 A1	12/2014	Bray et al.		GB	2369730 A	5/2002
2015/0062892 A1	3/2015	Krames et al.		JP	3-153212 A	7/1991
2015/0070337 A1	3/2015	Bell et al.		JP	5-18767 A	1/1993
2015/0310313 A1	10/2015	Murayama et al.		JP	8193727 A	7/1996
2015/0319882 A1	11/2015	Dunn et al.		JP	8-338981 A	12/1996
2015/0346525 A1	12/2015	Wolf et al.		JP	11-160727 A	6/1999
2015/0348460 A1	12/2015	Cox et al.		JP	2000122575 A	4/2000
2016/0037606 A1	2/2016	Dunn et al.		JP	2004325629 A	11/2004
2016/0055671 A1	2/2016	Menozzi et al.		JP	2005-148490 A	6/2005
2016/0162297 A1	6/2016	Shao		JP	2005265922 A	9/2005
2016/0198545 A1	7/2016	Dunn et al.		JP	2005-338266 A	12/2005
2016/0293142 A1	10/2016	Bowden et al.		JP	2006-106345 A	4/2006
2016/0334811 A1	11/2016	Marten		JP	2006-145890 A	6/2006
2016/0335698 A1	11/2016	Jones et al.		JP	2006318733 A	11/2006
2016/0338181 A1	11/2016	Schuch et al.		JP	2007003638 A	1/2007
2016/0338182 A1	11/2016	Schuch et al.		JP	2007322718 A	12/2007
2016/0358530 A1	12/2016	Schuch et al.		JP	2008-34841 A	2/2008
2016/0358538 A1	12/2016	Schuch et al.		JP	2008-83290 A	4/2008
2016/0360167 A1	12/2016	Mitchell et al.		JP	2008122695 A	5/2008
2017/0060369 A1	3/2017	Goyal et al.		JP	2009031622 A	2/2009
2017/0111486 A1	4/2017	Bowers et al.		JP	2010-181487 A	8/2010
2017/0111520 A1	4/2017	Bowers et al.		JP	2010-282109 A	12/2010
2017/0153473 A1 *	6/2017	Lee	G02F 1/1368	JP	2011-59543 A	3/2011
2017/0160775 A1	6/2017	Sun et al.		JP	2014-149485 A	8/2014
2017/0168295 A1	6/2017	Iwami		JP	2018-523148 A	8/2018
2018/0012565 A1	1/2018	Dunn		JP	2018-525650 A	9/2018
2018/0040297 A1	2/2018	Dunn et al.		KR	10-2006-0016469 A	2/2006
2018/0042134 A1	2/2018	Dunn et al.		KR	10-0768584 B1	10/2007
2018/0088368 A1	3/2018	Notoshi et al.		KR	10-2008-0000144 A	1/2008
2018/0129461 A1	5/2018	Kim-Whitty		KR	10-2008-0013592 A	2/2008
2018/0130385 A1	5/2018	Qian et al.		KR	10-2008-0086245 A	9/2008
2018/0132327 A1	5/2018	Dunn et al.		KR	10-2009-0014903 A	2/2009
2018/0203475 A1	7/2018	Van Derven et al.		KR	10-2010-0019246 A	2/2010
2018/0206316 A1	7/2018	Schuch et al.		KR	10-2011-0125249 A	11/2011
2019/0021189 A1	1/2019	Kim et al.		KR	10-2014-0054747 A	5/2014
2019/0116719 A1	4/2019	Fletcher et al.		KR	20140081903 A	7/2014
2019/0237045 A1	8/2019	Dunn et al.		KR	10-1759265 B1	7/2017
2019/0289754 A1	9/2019	Hubbard		KR	10-1931733 B1	12/2018
2019/0339312 A1	11/2019	Dunn et al.		KR	10-2047433 B1	11/2019
2019/0383778 A1	12/2019	Dunn et al.		KR	10-2130667 B1	6/2020
2020/0012116 A1	1/2020	Fuerst et al.		WO	2008/050402 A1	5/2008
2020/0033017 A1	1/2020	Brown et al.		WO	2010/141739 A2	12/2010
2020/0128701 A1	4/2020	Whitehead et al.		WO	2011/052331 A1	5/2011
2020/0150162 A1	5/2020	Dunn et al.		WO	2011/130461 A2	10/2011
2020/0211505 A1	7/2020	Dunn		WO	2011/150078 A2	12/2011
2020/0241612 A1 *	7/2020	Moon	H05K 7/20336	WO	2013/044245 A1	3/2013
2020/0294401 A1	9/2020	Kerecsen		WO	2016/183576 A1	11/2016
2020/0378939 A1	12/2020	Dunn et al.		WO	2017/031237 A1	2/2017
2020/0390009 A1 *	12/2020	Whitehead	G09F 9/3026	WO	2017/210317 A1	12/2017
2021/0034101 A1	2/2021	Yildiz et al.		WO	2018/009917 A1	1/2018
2021/0035494 A1	2/2021	Yildiz et al.		WO	2019/241546 A1	12/2019
2021/0263082 A1	8/2021	Dunn et al.				
2021/0302779 A1	9/2021	Dunn				
2021/0307214 A1	9/2021	Zhang et al.				
2022/0121255 A1	4/2022	Wang et al.				
2022/0187266 A1	6/2022	Dunn et al.				
2022/0295666 A1	9/2022	Dunn et al.				
2023/0060966 A1	3/2023	Dunn				

(56)

References Cited

FOREIGN PATENT DOCUMENTS

WO	2020/081687	A1	4/2020
WO	2022/197617	A1	9/2022

OTHER PUBLICATIONS

University of Miami, Heat Index and Dewpoint Climatology for Miami, FL, <https://bmcnoldy.rsmas.miami.edu/mia/>, URL accessed Apr. 18, 2022, 4 pages.

Novitsky, T. et al., Design How-To, Driving LEDs versus CCFLs for LCD backlighting, EE Times, Nov. 12, 2007, 6 pages, AspenCore.

Vogler, A. et al., Photochemistry and Beer, Journal of Chemical Education, Jan. 1982, pp. 25-27, vol. 59, No. 1.

Zeeff, T.M. et al., Abstract of EMC analysis of 18" LCD Monitor, Electromagnetic Compatibility, IEEE International Symposium, Aug. 21-25, 2000, vol. 1, 1 page.

Lee, X., What is Gamma Correction in Images and Videos?, http://xahlee.info/img/what_is_gamma_correction.html, Feb. 24, 2010, 4 pages.

Hooper, S. et al., Designing Mobile Interfaces, 2012, pp. 519-521, O'Reilly Media.

Outdoorlink, Inc., SmartLink Website User Manual, <http://smartlink.outdoorlinkinc.com/docs/SmartLinkWebsiteUserManual.pdf>, 2017, 33 pages.

Outdoorlink, Inc., SmartLink One, One Relay, <http://smartlinkcontrol.com/billboard/one-relay/>, retrieved Apr. 17, 2019, 2007-16, 6 pages.

Outdoorlink, Inc., SmartLink One Out of Home Media Controller, 2016, 1 page.

Rouaissia, C., Adding Proximity Detection to a Standard Analog-Resistive Touchscreen, SID 2012 Digest, 2012, 1564-1566, p. 132.

* cited by examiner

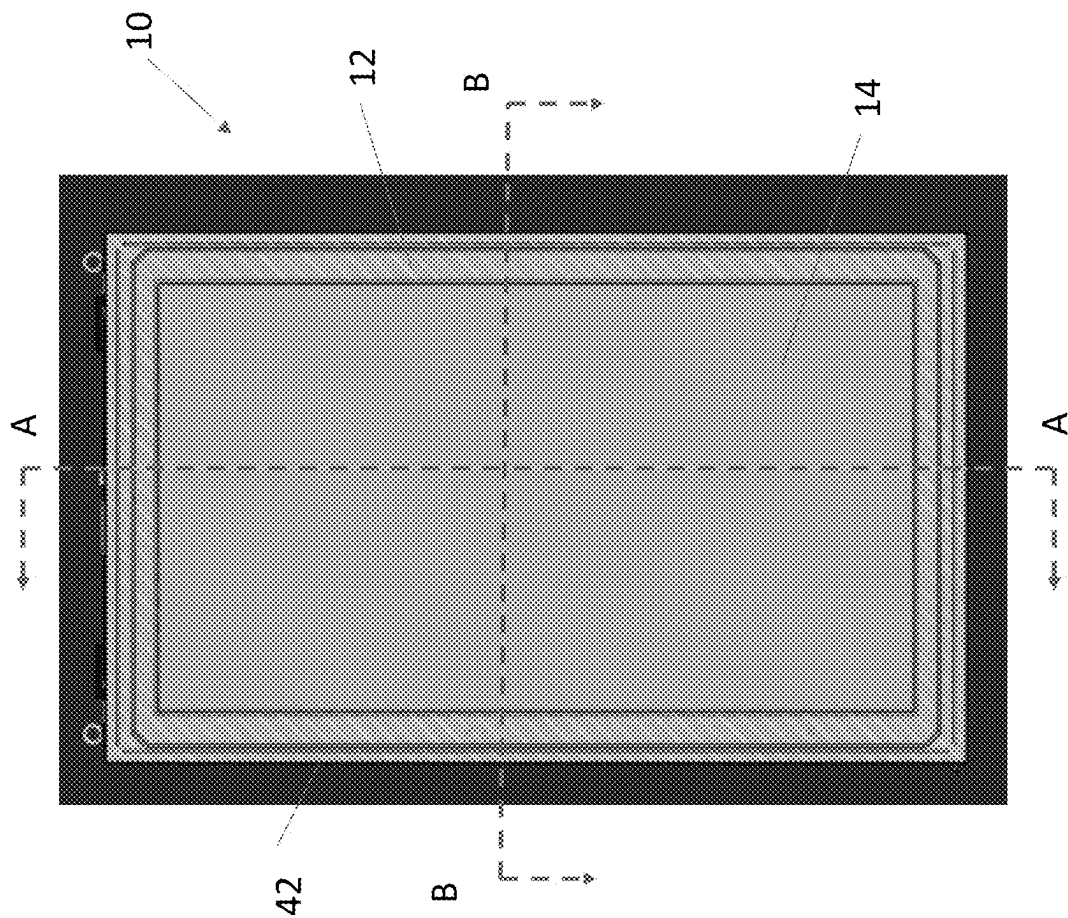


Figure 1

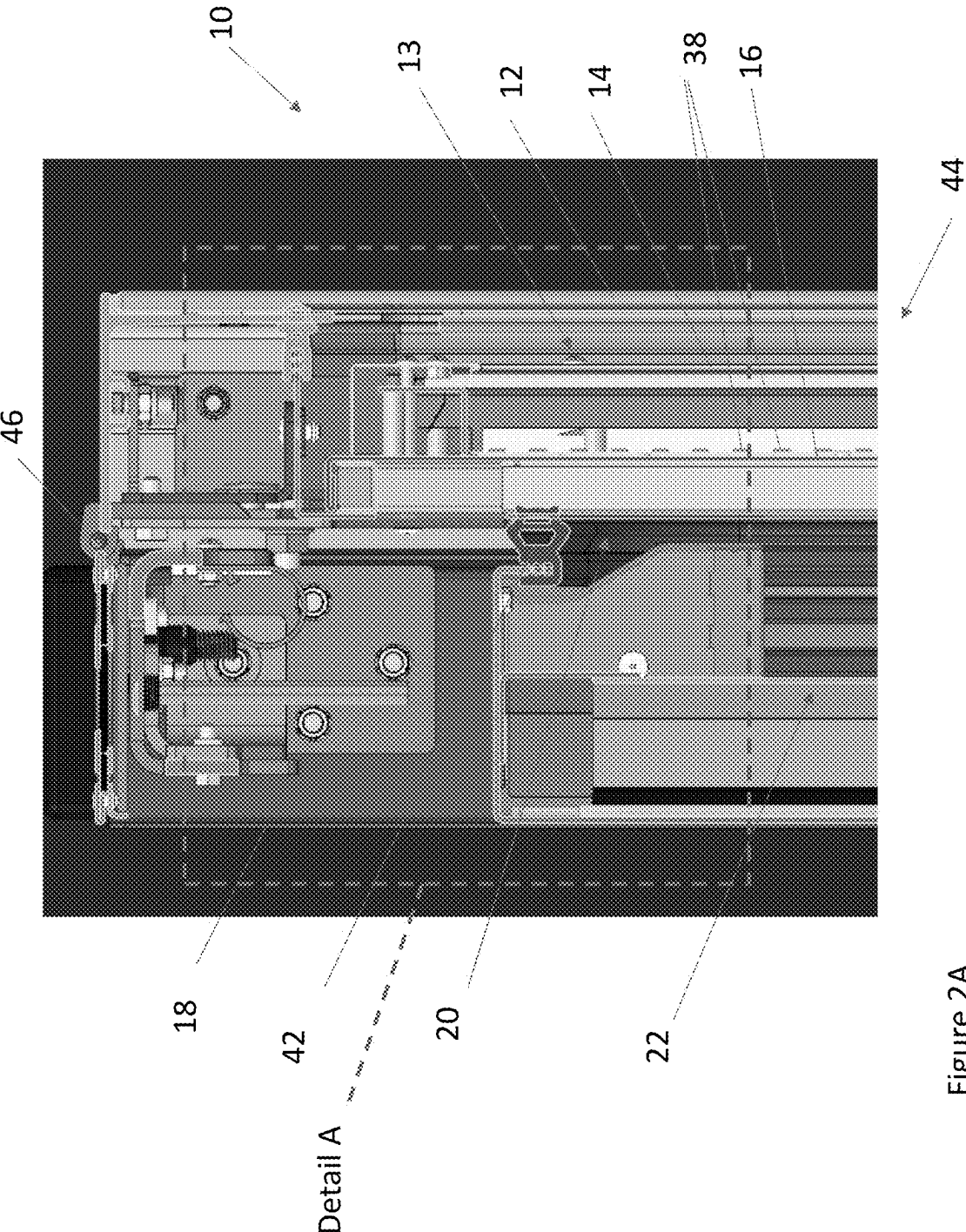


Figure 2A

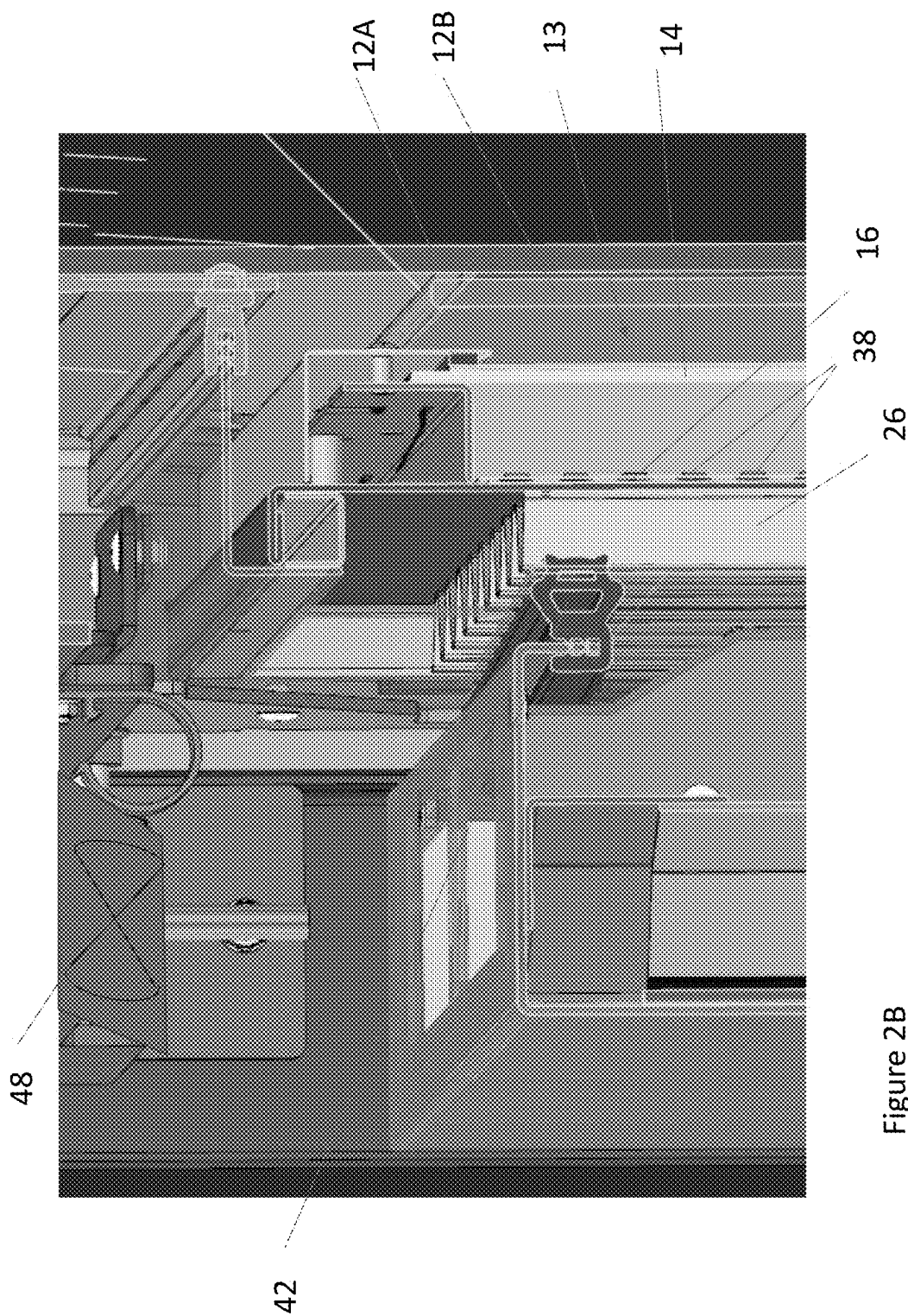


Figure 2B

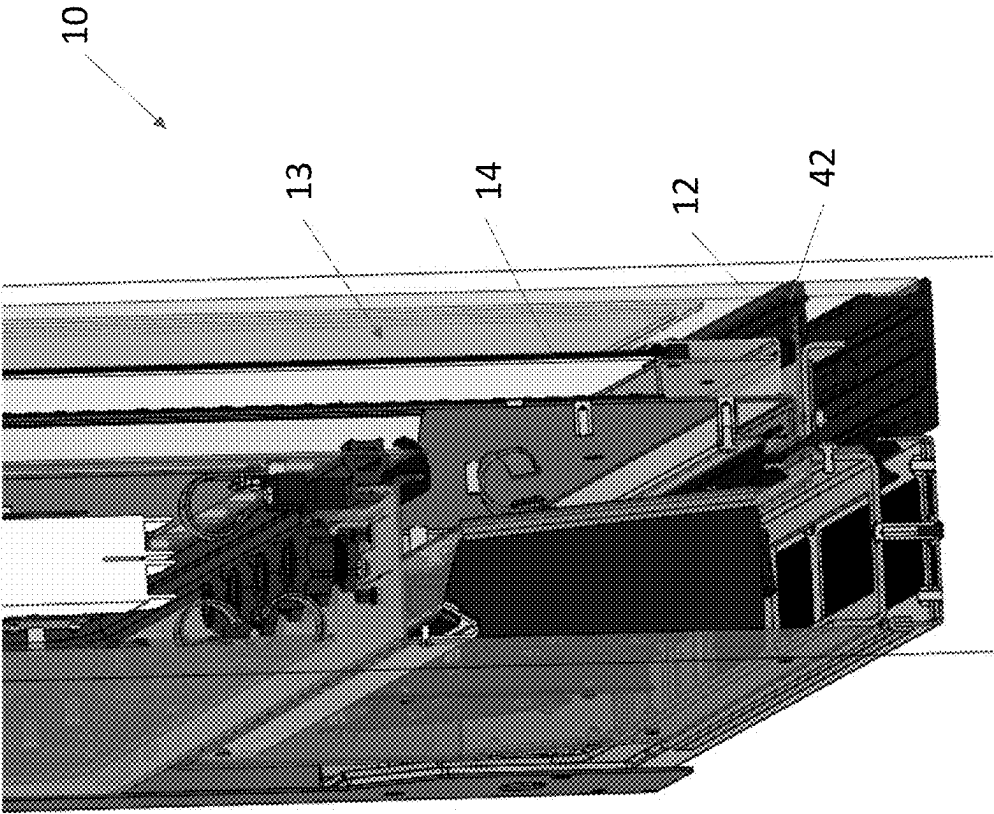


Figure 2C

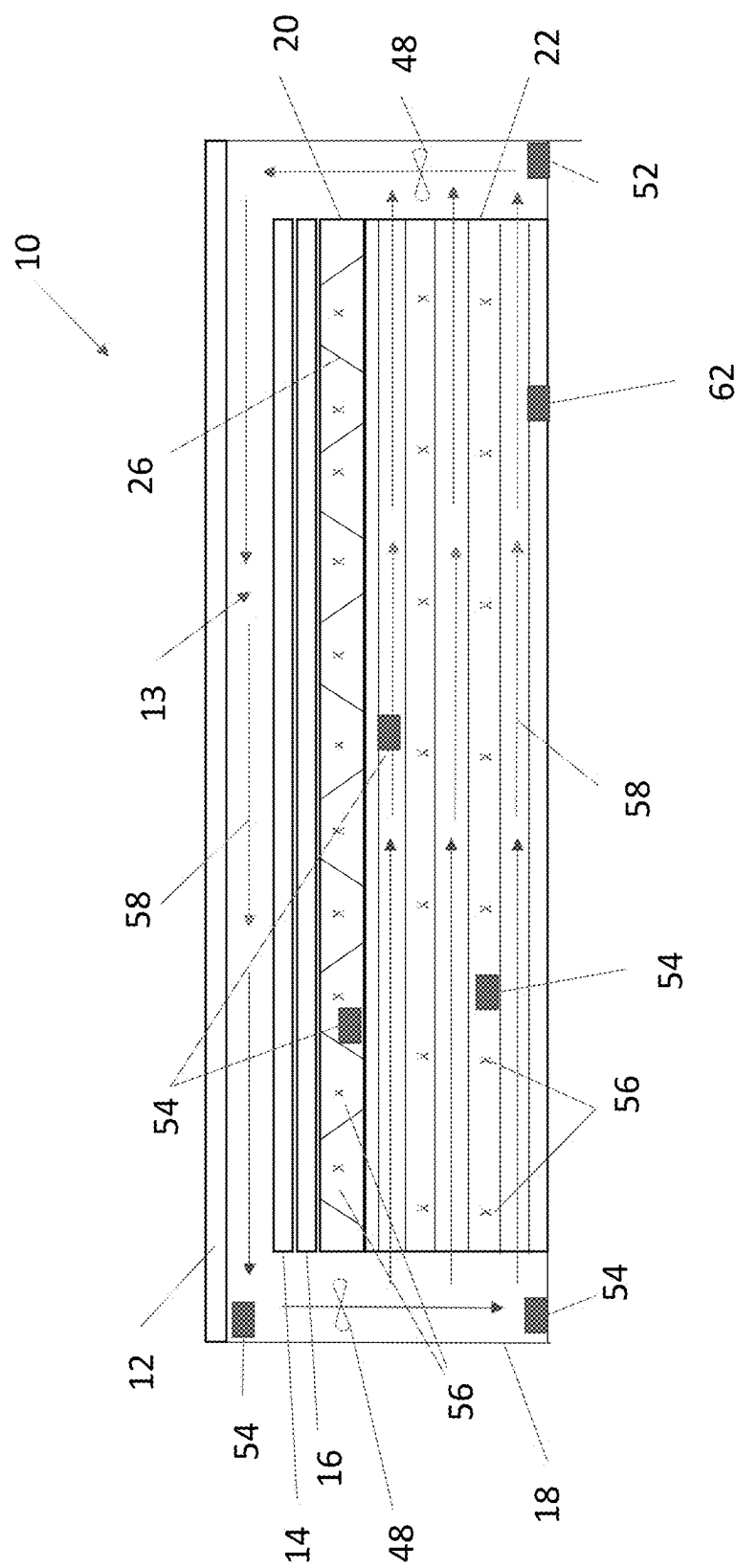


Figure 3

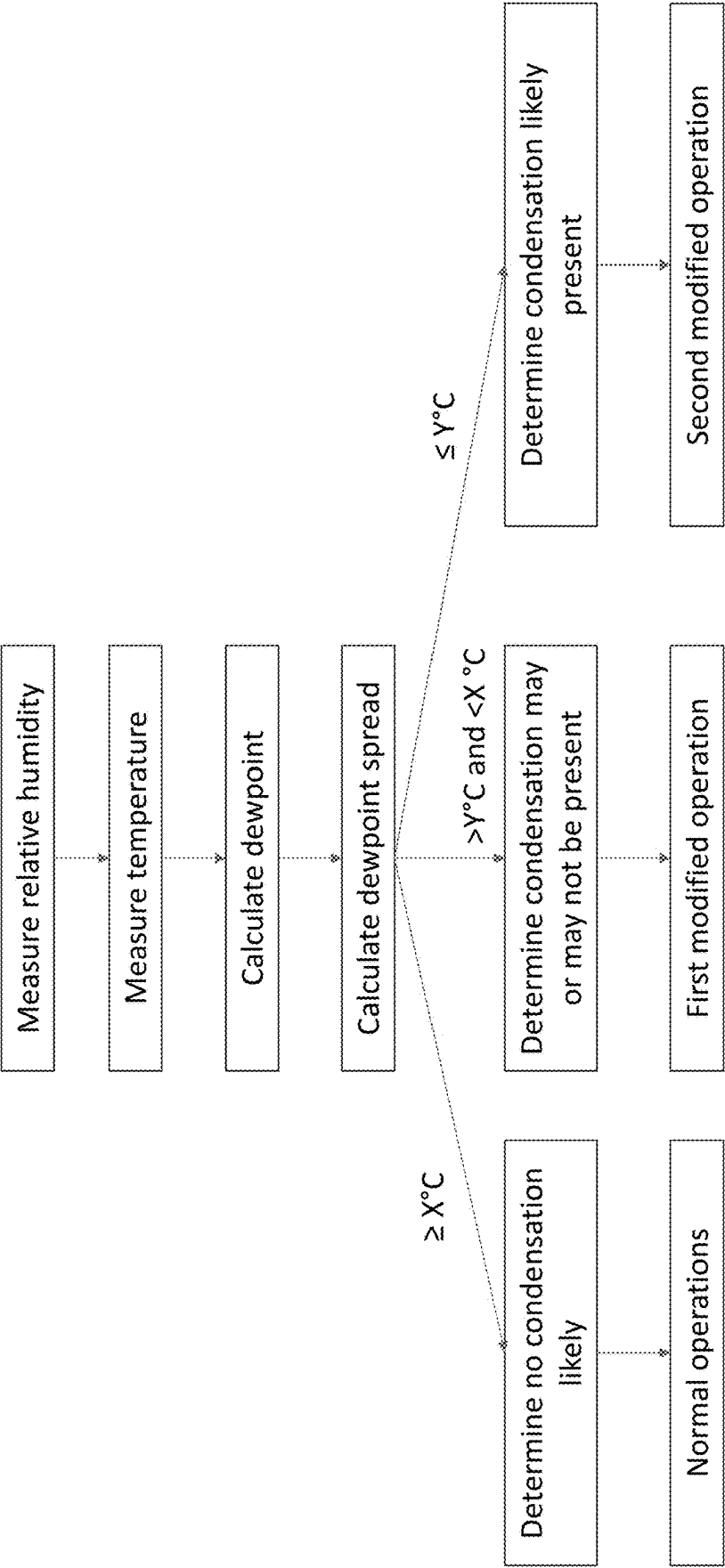


Figure 4

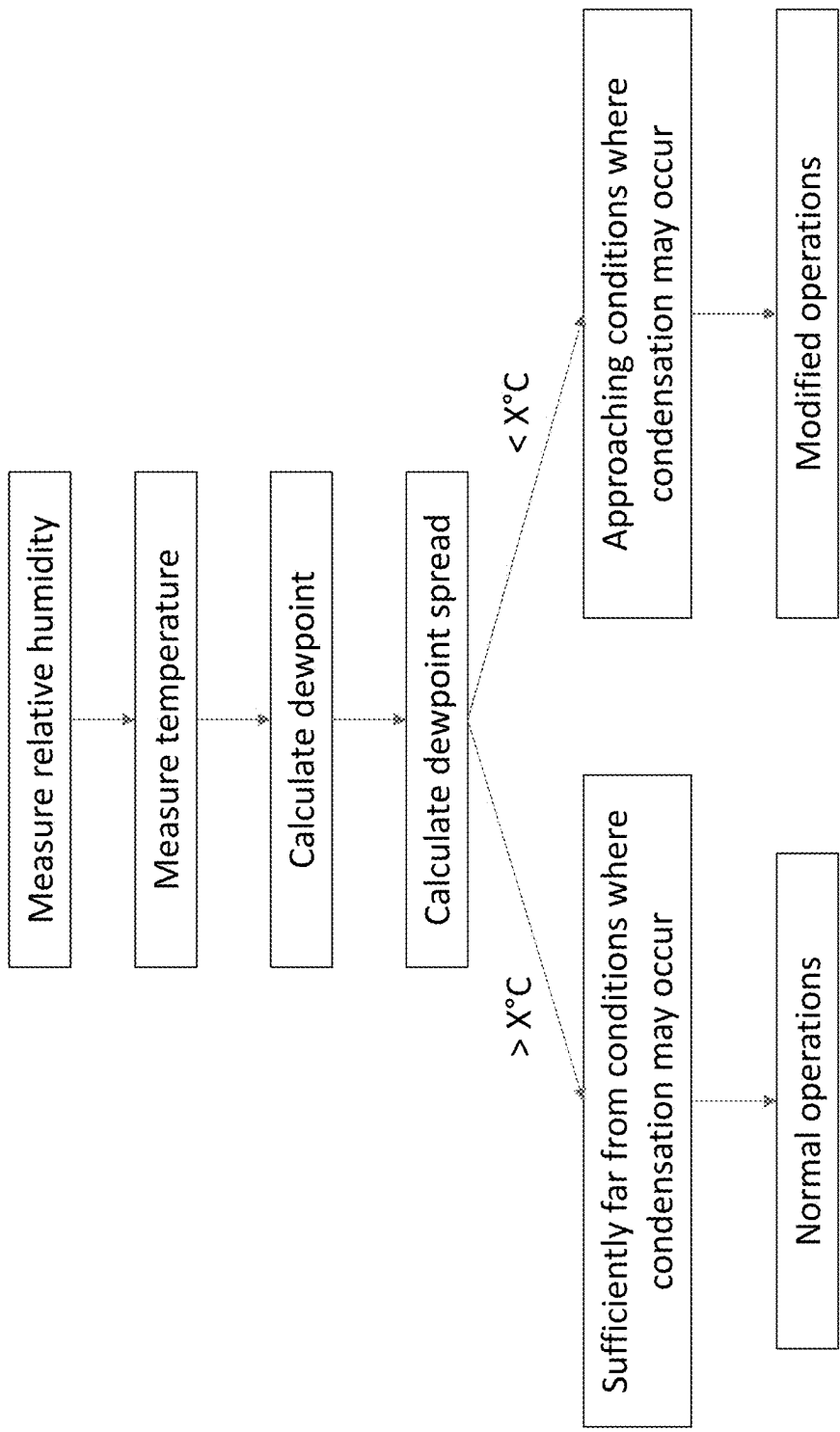


Figure 4B

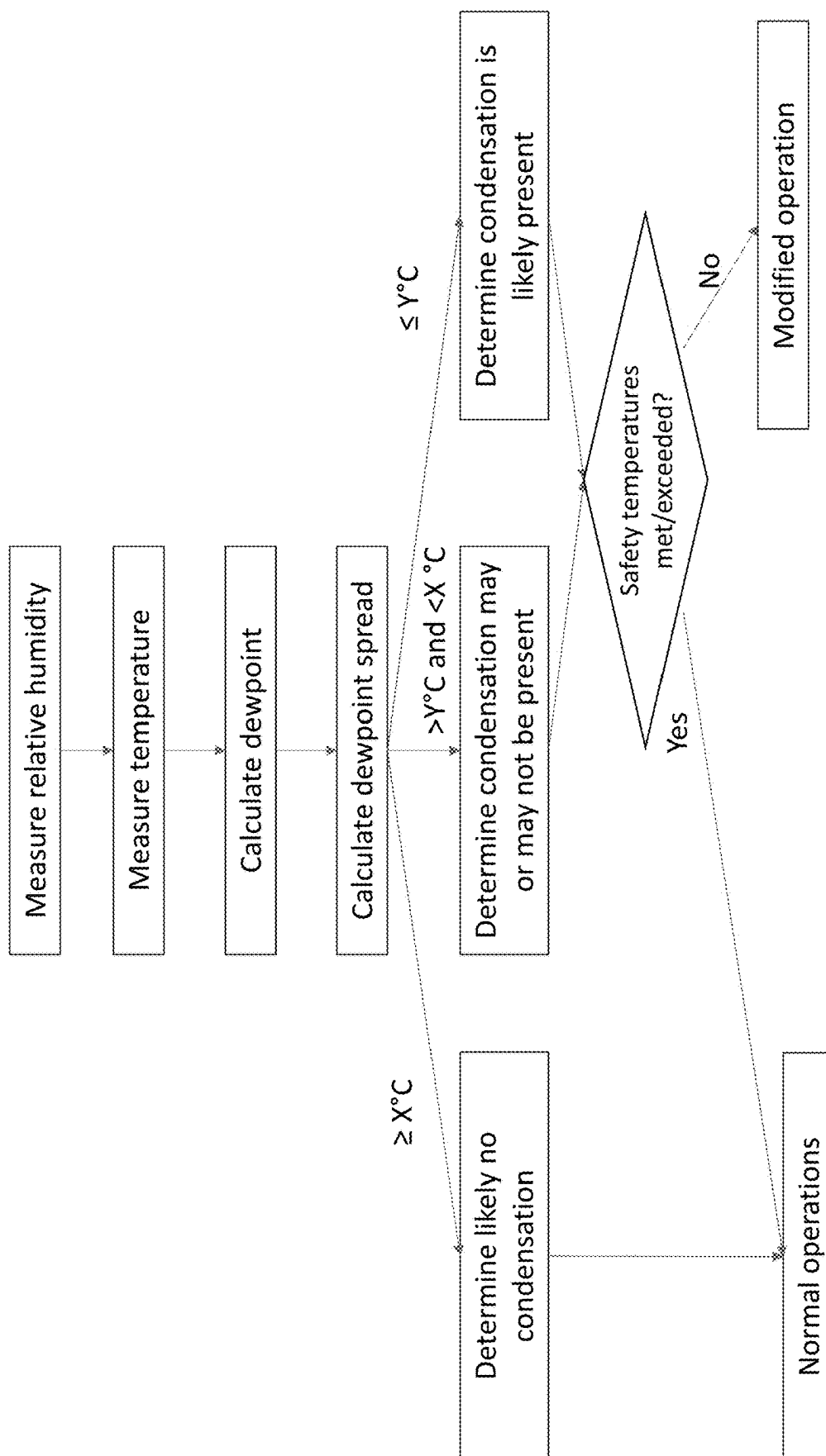


Figure 5

Humidity Zone Control Settings

70

Dewpoint Calculation Temp Source: ▾

Humidity Temp Sensor

16

68

Closed Loop Night Min

45.0

Closed Loop Temp Safety

48.0

Backlight Max Temp Safety

55.0

Power Supply Temp Safety

69.0

FPGA Die Temp Safety

51.0

Processor Temp Safety

41.0

72

Humidity Sensor Temp Safety

2.0

74

Dewpoint Spread Set Buffer

4.0

63

Dewpoint Spread Clear Buffer

120

Submit

64

Figure 6

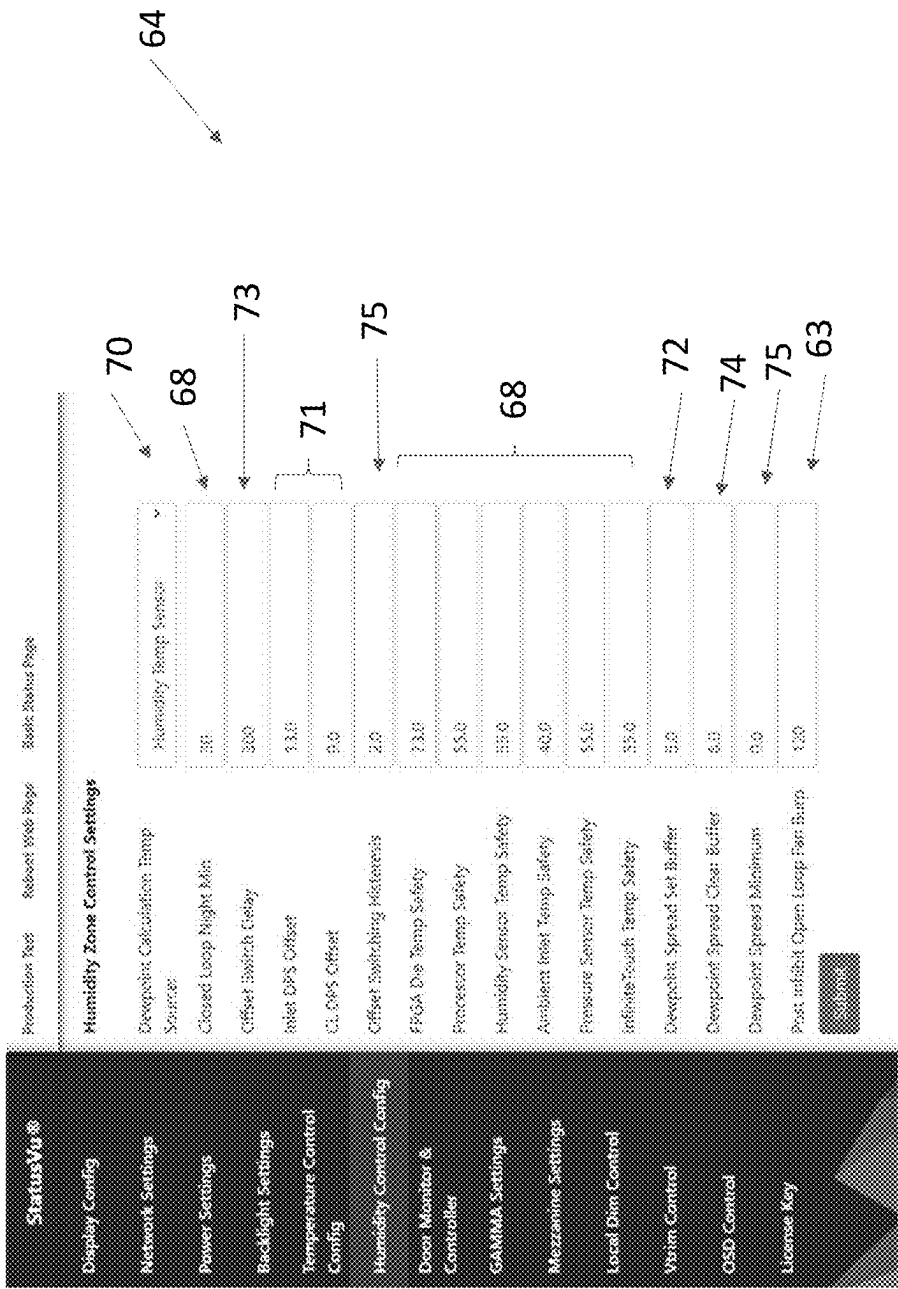


Figure 6B

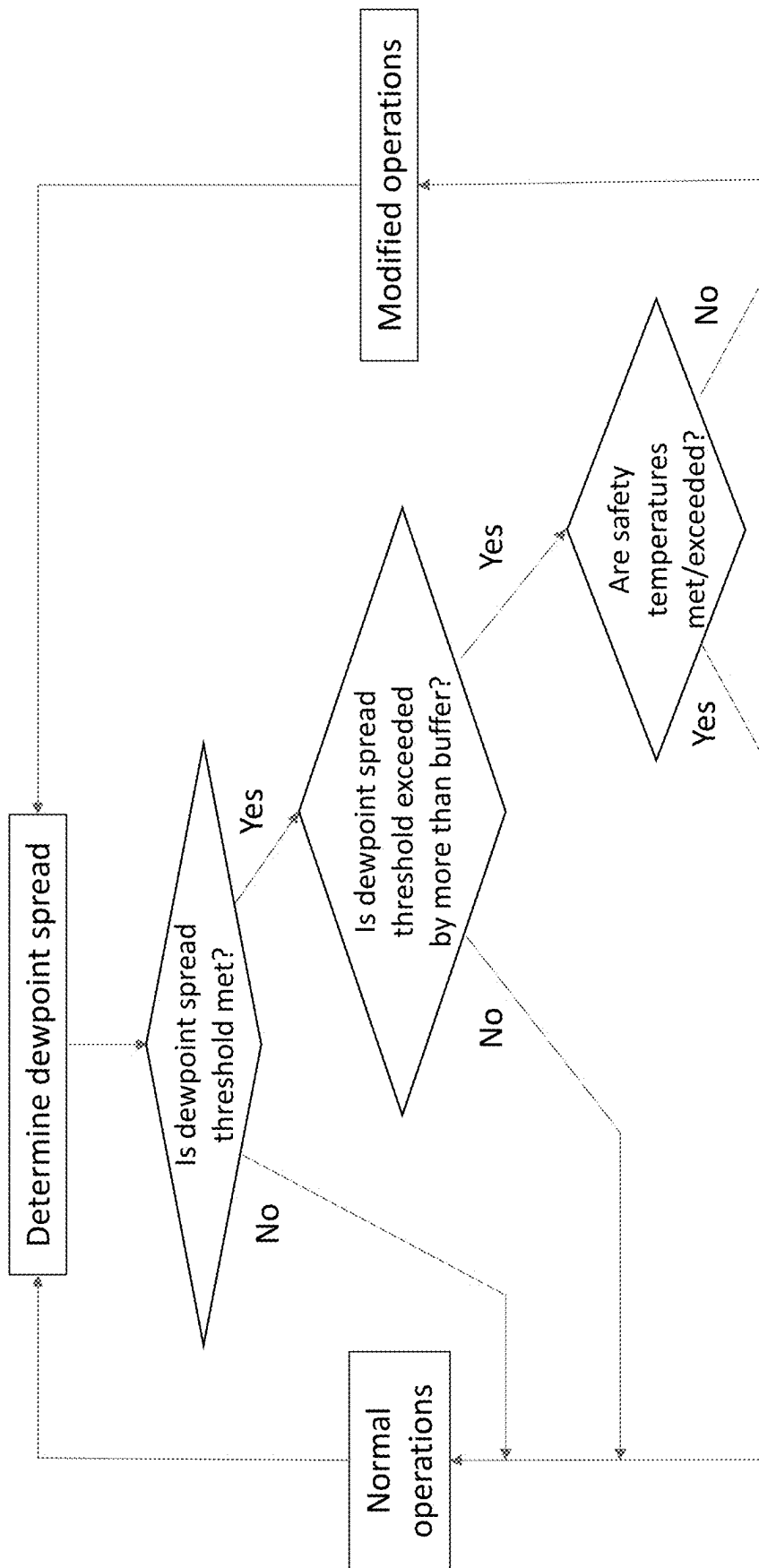


Figure 7A

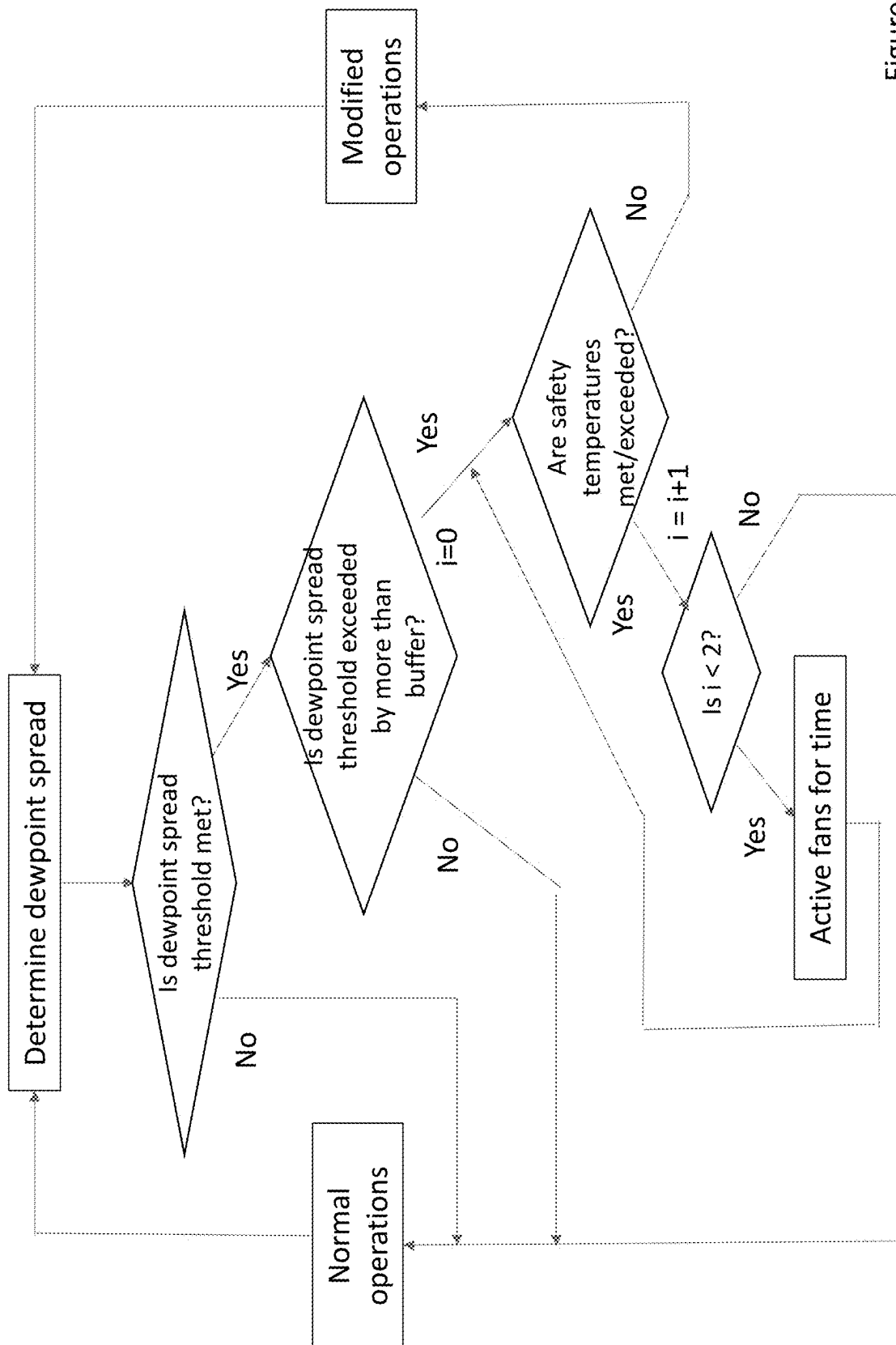


Figure 7B

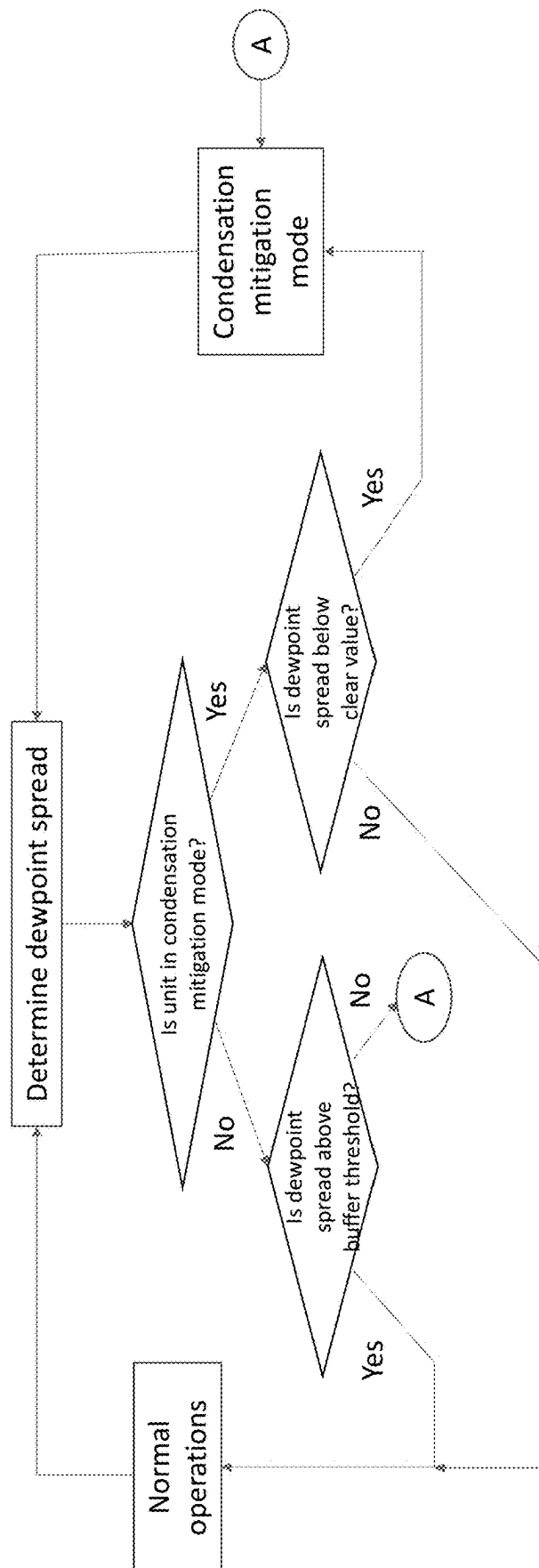


Figure 7C

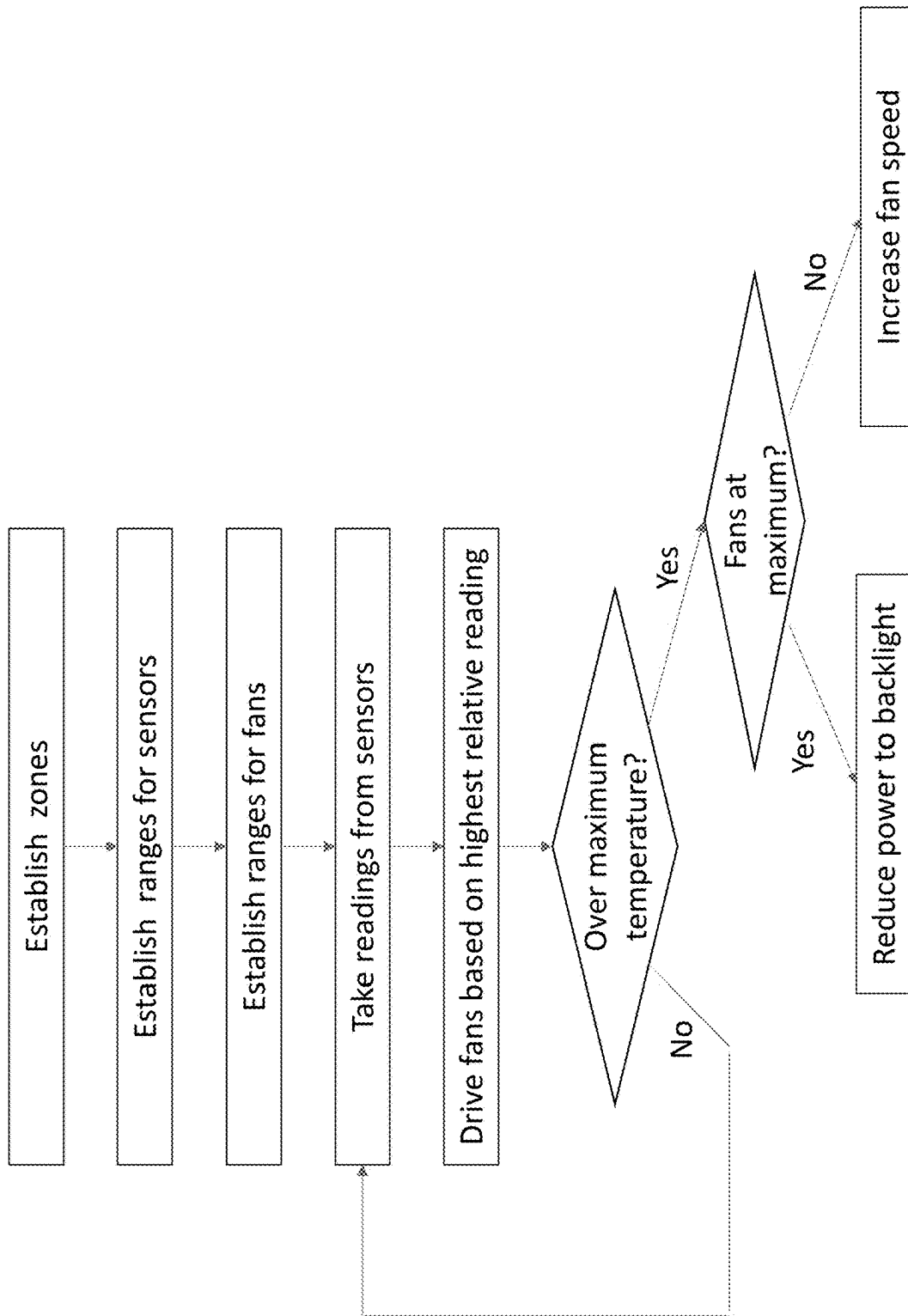


Figure 8

DISPLAY ASSEMBLIES WITH CONDENSATION MITIGATION AND RELATED SYSTEMS AND METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 18/213,709 filed Jun. 23, 2023, which is a continuation-in-part of U.S. application Ser. No. 17/694,261 filed Mar. 14, 2022, now U.S. Pat. No. 12,022,635 issued Jun. 25, 2024, which claims the benefit of U.S. provisional application Ser. No. 63/161,147 filed Mar. 15, 2021, and also claims the benefit of U.S. provisional application Ser. No. 63/239,273 filed Aug. 31, 2021, the disclosures of each of the foregoing are hereby incorporated by reference as if fully restated herein.

TECHNICAL FIELD

Exemplary embodiments relate generally to display assemblies with condensation mitigation features and related systems and methods.

BACKGROUND AND SUMMARY OF THE INVENTION

The use of electronic displays, such as for advertising, in the out-of-home market has increased in popularity over recent years. Being located outdoors, such electronic displays are frequently exposed to harsh conditions, including, but not limited to, solar loading, extreme temperatures, precipitation, moisture, contaminants, vandalism, wildlife, and the like. To protect the electronic displays and associated sensitive components from such harsh conditions, it is known to place the electronic displays in ruggedized housings. Such housings may fully or partially seal the electronic displays and other associated sensitive components.

It is known to thermally manage such electronic display assemblies using ambient air and/or circulating gas. Such ambient air may pass through one or more open loop airflow pathways within the assembly, and may thermally interact with circulating gas in one or more closed loop airflow pathways within the assembly where such closed loop pathways are used.

Operating such display assemblies in certain environments may result in the introduction of ambient air, such as into the one or more open loop airflow pathways, having a sufficiently different temperature relative to circulating gas and/or components of the display assembly as to result in the formation of condensation inside such display assemblies. For example, without limitation, the introduction of relatively cool ambient air into the display assembly may result in a sufficiently low dewpoint within the display assembly that water vapor in the ambient air and/or circulating gas within the assembly condenses into liquid, which may cause fogging and/or undesirable moisture exposure to sensitive electronic components. More specifically, for example, without limitation, the introduction of relatively cool ambient air into air-to-air heat exchangers contained within the display assembly may cause surfaces of these heat exchangers to drop below the dewpoint of the relative humidity contained within the fully or partially sealed enclosure resulting in condensation. Additionally, any inside surface of the fully or partially sealed enclosure may reach, or even drop below, the outside ambient temperature (e.g., when ice/snow is piled on top of a housing of the assembly),

resulting in cold spots within the assembly whose temperature is below the internal dewpoint which may cause the formation of condensation.

Furthermore, owners, operators, and/or manufacturers of such electronic display assemblies are increasingly undertaking power efficiency efforts. Such power efficiency efforts may include, for example, without limitation, decreasing illumination levels for lighting elements of the electronic display assemblies at night (e.g., partially or to zero), which may result in the electronic display assemblies becoming relatively cool, increasing the likelihood of condensation by lowering the dewpoint inside the display assembly. This may be particularly prevalent when combined with the ingestion of relatively cool ambient air.

What is needed are systems and methods for controlling condensation in electronic display assemblies. Systems and methods for controlling condensation in electronic display assemblies are provided.

In general, gaskets utilized in such electronic display assemblies may be sufficient to entirely or substantially keep out liquids, but sometimes such gaskets are not gas-tight or entirely gas-tight. Therefore, moisture can sometimes permeate through the gasket, such as in the form of water vapor, and enter an otherwise closed loop airflow pathway. Such closed loop airflow pathways may still be considered sealed, as such closed loop airflow pathways are kept entirely or substantially free from solid or liquid particulate such as, but not limited to, dust, debris, precipitation, combinations thereof, or the like. In general, when the interior of an otherwise fully or partially sealed enclosure (e.g., closed loop airflow pathway) is warmer than the outdoor environment, moisture may escape the otherwise fully or partially sealed enclosure, such as, but not limited to, by way of gaseous particles in the air which may permeate the liquid-tight, but not necessarily vapor-tight, gaskets. More specifically, for example, without limitation, as heat is added to the circulating gas within the closed loop airflow pathways, the circulating gas may expand and some of that expanding air may be forced through the gasket or otherwise to a location outside of the closed loop airflow pathway, and take moisture in the circulating gas with it. The converse may also be true.

The relative humidity within an electronic display assembly may be determined, such as, but not limited to, by way of one or more sensors configured to measure humidity and/or temperature. The relative humidity may be determined at the one or more sensors or at a separate controller. The same or different temperature measures may be used in conjunction with the relative humidity to determine the dewpoint of air inside the electronic display assembly. The measurements from multiple sensors may be used and/or multiple readings may be aggregated in various ways.

A dewpoint spread may be calculated between the temperature of ambient air and the calculated dewpoint. For example, without limitation, the temperature of the ambient air, such as for calculating the dewpoint and/or dewpoint spread, may be determined as the lesser of one or more temperature readings of ambient air at said intake or along said portion of said one or more open loop airflow pathways of the assembly or an internet-retrieved local ambient air temperature.

Where the dewpoint spread is determined to be less than 0° C., it may be determined with a high degree of confidence that condensation is occurring. Where the dewpoint spread is determined to be greater than 0° C., but less than a predetermined threshold "X", which may be in the range of +2 to +5° C. in exemplary embodiments, it may be deter-

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mined that condensation may be occurring. Where the dewpoint spread is determined to be greater than $X^{\circ}\text{C.}$, it may be determined with a high degree of confidence that condensation will not occur. Modified operations may be undertaken where it is determined that condensation is occurring or may be occurring according to the above-described criteria, but not when condensation is not occurring according to the above-described criteria. Determining which of the interior surfaces is coldest, and thus potentially having condensation, may be difficult to determine. Thus, modified operations may be conservatively undertaken for both where condensation is occurring and where condensation may be occurring according to the above-described criteria.

In exemplary embodiments, without limitation, where the dewpoint spread is greater than or equal to $X^{\circ}\text{C.}$, which is variable but in exemplary embodiments may be 2°C. , between 2°C. - 5°C. , 4°C. , or 5°C. for example, without limitation, a determination may be made that no condensation is likely. Where the dewpoint spread is greater than $Y^{\circ}\text{C.}$, which is variable but in exemplary embodiments may be 0°C. , between 0°C. - 2°C. , or 2°C. for example, without limitation, and less than $X^{\circ}\text{C.}$, a determination may be made that condensation may or may not be present. Where the dewpoint spread is less than or equal to $Y^{\circ}\text{C.}$, a determination may be made that condensation is likely or definitely present within the electronic display assembly. In exemplary embodiments, the controller may be configured to operate the assembly normally (e.g., without any modified operations to minimize, reduce, control, and/or eliminate condensation formation within the electronic display assembly) where the determination is made that no condensation is likely, and may initiate certain modified operations where the determination is made that condensation may or may not be present and/or that condensation is likely present. Such modified operations may be configured to minimize, reduce, control, and/or eliminate condensation formation within the electronic display assembly.

Alternatively, or additionally, a single dewpoint spread threshold may be utilized such that the units are programmed to operate in either a normal operating mode or a condensation mitigation mode. For example, without limitation, where the dewpoint spread is greater than "X" the unit may operate normally, but where the dewpoint spread is less than "X" the unit may operate in condensation mitigation mode. When equal to "X" the unit may be configured to either operate in the normal mode or the condensation mitigation mode. In exemplary embodiments, the threshold for activating condensation mitigation mode may be different from deactivating condensation mitigation mode. For example, the threshold for transitioning from normal operating mode to condensation mitigation mode may be $X-A$ (e.g., 3°C. by way of non-limiting example) while the threshold for transitioning from condensation mitigation mode to normal operating mode may be $X+B$ (e.g., 6°C. by way of non-limiting example). An electronic notification may be generated and transmitted where the dewpoint spread reaches a predetermined threshold below "X" (e.g., $X-C$, such as 0°C. by way of non-limiting example).

In other exemplary embodiments, where a determination is made that condensation may or may not be present and/or that condensation is likely present (e.g., dewpoint spread less than $X^{\circ}\text{C.}$, or less than or equal to $X^{\circ}\text{C.}$), a determination may be made as to whether certain safety thresholds are met and/or exceeded. If the safety thresholds are met and/or exceeded, the electronic display assembly may be configured to operate normally (e.g., without modified,

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condensation control operations). In this way, the operational safety of the display assembly and/or its components may be prioritized. If the safety thresholds are not met and/or exceeded, the electronic display assembly may operate under the modified operating parameters configured to minimize, control, reduce, and/or eliminate condensation within the electronic display assembly.

In yet other exemplary embodiments, modified operations may be undertaken based on a dewpoint threshold and buffer, with or without the secondary check for safety thresholds. For example, without limitation, a threshold dewpoint spread of $A^{\circ}\text{C.}$ and a buffer of $B^{\circ}\text{C.}$ may be set. Where the dewpoint spread exceeds $A^{\circ}\text{C.}$ by more than $B^{\circ}\text{C.}$, modified operations may be undertaken. In exemplary embodiments, a check to ensure that the safety thresholds are not met or exceeded may be first undertaken. $A^{\circ}\text{C.}$ and/or $B^{\circ}\text{C.}$ may each be independent variables and may be, for example, without limitations, each 2°C. , $A=4^{\circ}\text{C.}$ and $B=2^{\circ}\text{C.}$, $A=5^{\circ}\text{C.}$ and $B=1-4^{\circ}\text{C.}$, combinations thereof, or the like.

The safety thresholds may comprise any temperature or other condition of any components. For example, without limitation, the safety thresholds may be designed to prevent overheating of sensitive, critical, and/or expensive electronic components. Modified operations, as explained more fully below, may reduce or eliminate ambient air introduction and/or promote heat generation within the unit. The normal operations may permit the ingestion of increased or unlimited amounts of ambient air, which may be prioritized over condensation where the safety thresholds are met and/or exceeded. Where such safety thresholds are not met and/or exceeded, condensation control may be prioritized.

Alternatively, or additionally, the safety thresholds may not be required.

Modified operations may include, for example, without limitation, increasing heat in the electronic display assembly, such as, but not limited to, by changing, such as by reducing, fan speed, operating time, combinations thereof, or the like and/or increasing power to lighting elements, restricting the ability to turn off or reduce power to the lighting elements, combination thereof, or the like. Modified operations may reduce, prevent, control, and/or eliminate the formation of condensation within the display assembly by decreasing the ingestion of outside (ambient) air, which may be relatively cold (e.g., turning ambient air fans off and/or minimizing their operation) and/or increasing power to lighting elements (e.g., turning on, increasing power, preventing dimming).

For non-emissive displays, such as LCDs, overall image luminance may be kept low as desired (e.g., no noticeable increase in visible image luminance) by turning backlight power up while concurrently turning image gray scale down, for example, without limitation. For non-emissive or emissive displays, power consumption may alternatively, or additionally, be increased by turning drive current to maximum while turning pulse width modulation (PWM) control to minimum levels required to maintain desired perceptive brightness, for example, without limitation. In exemplary embodiments, modified operations may be performed once internal and/or ambient temperatures, dewpoint, relative humidity, and/or dewpoint spread reach a certain threshold, range, combination thereof, or the like and/or where certain safety thresholds are not yet met and/or exceeded.

Such condensation controls may be particularly useful where the electronic display assembly is powered down or otherwise placed in a reduced power mode and/or when ambient temperatures drop, one or both of which may occur

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during nighttime hours, winter hours, and/or under power efficiency efforts to name a few examples. In exemplary embodiments, without limitation, the systems and methods shown and/or described herein may accomplish condensation control without the need for a separate and/or dedicated heater, thereby reducing power consumption and/or noise.

The various criteria described herein, including, but not limited to, the dewpoint spread ranges and safety thresholds, are merely exemplary and are not intended to be limiting. Other criteria and/or thresholds may be utilized. For example, without limitation, other data points, dewpoint spread criteria, operation modifications, temperatures, thresholds, safety thresholds, and/or ranges, may be utilized.

Alternatively, or additionally, it may be desirable to operate air circulation devices within the display assembly to control temperatures within the display assembly and/or provide condensation control. In exemplary embodiments, the air circulation devices may comprise fan units, each of which may comprise one or more fans, and may be associated with one or more sensors, such as temperature sensors. Zones may be virtually defined within the display assemblies, each of which may include one or more of the air circulation devices and one or more of the associated sensors.

Operational ranges for the air circulation devices may be established. Such operational ranges may be programmed at, or stored at, the controller(s). Such operational ranges may be stored in association with one or more of the air circulation devices. Desired operating ranges may be established for the sensors. Such desired operating ranges may be programmed at, or stored at, the controller(s). Such desired operating ranges may be stored in association with one or more of the sensors.

Operational ranges for the air circulation devices and/or desired operating ranges for the sensors may be specific to the date, time, ambient conditions, combinations thereof, or the like. Operational ranges for the air circulation devices and/or desired operating ranges for the sensors may be specific to the zone, air circulation device, and/or sensor or for the whole display assembly.

Readings from the sensors may be taken periodically, continuously, sporadically, or the like. Operation of some or all of the air circulation devices may be controlled by the highest sensor reading relative to the associated desired operating range. Such control may be performed on a zone-by-zone basis or for the entire display assembly.

Where a maximum operating temperature is reached or exceeded at one or more of the sensors, speed or other operating conditions (e.g., number of active fans, volumetric flow rate, power supplied, etc.) of the air circulation devices may be increased, such as by the controller(s), until a maximum operational level is reached. If the maximum operating temperature is reached or exceeded at one or more of the sensors, power to the backlight may be reduced until temperatures fall below the maximum operating temperatures. Such reduction may be made in an inversely proportional fashion to how far the temperature has exceeded the maximum operating temperature.

Sensor readings may be continuously or periodically retaken and operations adjusted accordingly.

One or more gaskets or other sealing devices for separating an open loop and/or ambient environment from a closed loop or at least partially sealed area may be provided between a portion of the housing for the unit (e.g., a framework or chassis) and a cover glass so as to reduce or prevent thermal transfer through the portion of the housing for the unit, which may comprise metal.

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Further features and advantages of the systems and methods disclosed herein, as well as the structure and operation of various aspects of the present disclosure, are described in detail below with reference to the accompanying figures.

BRIEF DESCRIPTION OF THE DRAWINGS

In addition to the features mentioned above, other aspects of the present invention will be readily apparent from the following descriptions of the drawings and exemplary embodiments, wherein like reference numerals across the several views refer to identical or equivalent features, and wherein:

FIG. 1 is a perspective view of an exemplary display assembly in accordance with the present invention and indicating section lines A-A and B-B;

FIG. 2A is a detailed side sectional view of the display assembly of FIG. 1 taken along section line A-A and indicating detail A;

FIG. 2B is a detailed perspective view of detail A of FIG. 2A;

FIG. 2C is a detailed side sectional view of another exemplary embodiment of the display assembly of FIG. 1 taken along section line A-A;

FIG. 3 is a simplified top sectional view of the display assembly of FIG. 1 taken along section line B-B;

FIG. 4 is a flow chart with exemplary logic for operating the electronic display assembly of FIGS. 1-3 in accordance with the present invention;

FIG. 4B is a flow chart with other exemplary logic for operating the electronic display assembly of FIGS. 1-3 in accordance with the present invention;

FIG. 5 is a flow chart with other exemplary logic for operating the electronic display assembly of FIGS. 1-3 in accordance with the present invention;

FIG. 6 is a plan view of an exemplary operations programming interface for the electronic display assemblies of FIGS. 1-5;

FIG. 6B is a plan view of another exemplary operations programming interface for the electronic display assemblies of FIGS. 1-5;

FIG. 7A is a flow chart with other exemplary logic for operating the electronic display assembly of FIGS. 1-6 in accordance with the present invention;

FIG. 7B is a flow chart with other exemplary logic for operating the electronic display assembly of FIGS. 1-7A in accordance with the present invention;

FIG. 7C is a flow chart with other exemplary logic for operating the electronic display assembly of FIGS. 1-6 in accordance with the present invention; and

FIG. 8 is a flow chart with exemplary logic for operating the electronic display assembly of FIGS. 1-7B in accordance with the present invention.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENT(S)

Various embodiments of the present invention will now be described in detail with reference to the accompanying drawings. In the following description, specific details such as detailed configurations and components are merely provided to assist the overall understanding of these embodiments of the present invention. Therefore, it should be apparent to those skilled in the art that various changes and modifications of the embodiments described herein can be made without departing from the scope and spirit of the

present invention. In addition, descriptions of well-known functions and constructions are omitted for clarity and conciseness.

Embodiments of the invention are described herein with reference to illustrations of idealized embodiments (and intermediate structures) of the invention. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments of the invention should not be construed as limited to the particular shapes of regions illustrated herein but are to include deviations in shapes that result, for example, from manufacturing.

FIG. 1 through FIG. 3 illustrate an exemplary electronic display assembly 10. The assembly 10 may comprise a housing 18. The assembly 10 may comprise a cover panel 12. The cover panel 12 may comprise a single layer or multiple layers 12A, 12B bonded together, such as by way of one or more optical adhesives. The cover panel 12 may be located forward of, and may be spaced apart from, an electronic display layer 14. The electronic display layer 14 may comprise a layer of liquid crystals (e.g., an LCD), a plasma display, OLED display, LED display, combinations thereof, or the like. The cover panel 12 may form a forward portion of the housing 18. The housing 18 may substantially enclose the electronic display layer 14.

The cover panel 12 may be transparent or translucent such that images displayed at the electronic display layer 14 are visible to an intended viewer through the cover panel 12. The cover panel 12 may be configured to protect the electronic display layer 14 and/or other components of the electronic display assembly 10. The cover panel 12 may, alternatively, or additionally, be configured to enhance optics of the images displayed at the electronic display layer 14. The cover panel 12 and/or electronic display layer 14 may comprise one or more polarizers, anti-reflective films, surface treatments, combinations thereof, or the like. A front air gap 13 may be located rearward of the cover panel 12 and forward of the electronic display layer 14. The front air gap 13 may form part of a closed loop airflow pathway for circulating gas 58.

An illumination device 16 may be provided adjacent to at least a portion of the electronic display layer 14. The illumination device 16 may comprise a number of lighting elements 38. The lighting elements 38 may comprise light emitting diodes (LEDs), though other kinds or types of lighting elements 38 may be utilized. The illumination device 16 may be configured to provide illumination to the electronic display layer 14 when powered. For example, without limitation, the illumination device 16 may be configured to provide direct backlight for the electronic display layer 14 and may be positioned rearward of the electronic display layer 14. Alternatively, or additionally, the illumination device 16 may be configured to provide edge lighting for the electronic display layer 14 and may be positioned around some or all of a perimeter of the electronic display layer 14, one or more light guides, reflective elements, combinations thereof, or the like. In exemplary embodiments, the illumination device 16 may comprise a number of the lighting elements 38 provided on one or more tiles, mounted to a substrate (e.g., printed circuit board), combinations thereof, or the like. Any number, arrangement, and/or type of the lighting elements 38 may be used.

In other exemplary embodiments, the electronic display layer 14 may be an emissive display and/or may be configured to illuminate without the need for a separate and/or dedicated illumination device 16. Examples of such embodi-

ments include, without limitation, OLED displays, plasma displays, LED displays, combinations thereof, or the like.

The assembly 10 may comprise one or more open loop heat exchangers (hereinafter also "OL HX") 20. The OL HX 20 may be configured to accommodate ambient air. The OL HX 20 may be provided rearward of the illumination device 16. In exemplary embodiments, the OL HX 20 may extend along some or all of the illumination device 16 so as to absorb some or all of the heat generated by the illumination device 16 when in use. The OL HX 20 may extend directly along the illumination device 16 or may be spaced apart therefrom. For example, without limitation, one or more thermally conductive layers, air gaps, and/or spacers may be positioned between the illumination structure and the OL HX 20.

The OL HX 20 may comprise one or more layers. In exemplary embodiments, some or all of the layers of the OL HX 20 may comprise a corrugated structure 26. The corrugated structure 26 may comprise a zigzag pattern which extends between two or more panels or layers of the OL HX 20, thereby forming a number of channels or pathways within the OL HX 20. Alternatively, or additionally, the OL HX 20 may comprise a number of tubes (e.g., square, rectangular, round, combinations thereof, or the like) defining passageways or channels for ambient air.

In exemplary embodiments, the OL HX 20 may be in fluid communication with one or more intakes and exhausts provided in the housing 18. Such intakes and exhausts may comprise one or more apertures in the housing 18 which permit the intake and exhaust, respectively, of ambient air from the assembly 10.

One or more air circulation devices 48, such as, but not limited to, fans may be provided within or otherwise in fluid communication with the OL HX 20 and/or other portions of one or more open loop airflow pathways within the electronic display assembly 10 to cause the ingestion of ambient air into the assembly 10, flow through the one or more open loop airflow pathways, and exhaustion from the assembly 10 when operated. Such air circulation devices 48 may be in electronic communication with one or more controller(s) 52. The air circulation devices 48 may comprise, for example, without limitation, axial fans, centrifugal fans, combinations thereof, or the like. Any number and/or type of air circulation devices 48 at any one or number of locations within the display assembly 10 may be utilized.

The assembly 10 may comprise one or more open loop/closed loop heat exchangers (hereinafter also "OL/CL HX") 22. The OL/CL HX 22 may be provided rearward of the illumination device 16. The OL/CL HX 22 may comprise multiple layers. In exemplary embodiments, each of the layers may be configured to accommodate one of: only ambient air as part of one or more of the open loop airflow pathways in the assembly 10, or circulating gas as part of one or more closed loop airflow pathways in the assembly 10. The layers may be arranged, for example, without limitation, vertically or horizontally adjacent one another. For example, without limitation, the layers may alternate between being configured to accommodate ambient air and circulating gas. In exemplary embodiments, a first portion of the layers may form part of the same or a different open loop airflow pathway as the OL HX 20 and a second portion of the layers may form part of the same or a different closed loop airflow pathway as the front air gap 13. The OL/CL HX 22 may be in fluid communication with the same or different intake(s) and exhaust(s) as the OL HX 20. The OL/CL HX 22 may be in fluid communication with the front air gap 13, though such is not required.

In exemplary embodiments, the electronic display assembly **10** may comprise multiple electronic display layers **14**. In such embodiments, the electronic display assembly **10** may comprise multiple cover panels **12**, illumination structures **16**, OL HX **20**, OL/CL HX **22**, combinations thereof, or the like. However, at least the OL/CL HX **22** may be common to multiple electronic display layers **14** in some embodiments. For example, without limitation, the electronic display assembly **10** may comprise a first and second electronic display layer **14** provided in a back-to-back arrangement with front air gaps **13** fluidly connected to a common OL/CL HX **22** but separate OL HXs **20** for each electronic display layer **14**. Such electronic display layers **14** may be provided in the same or different housings **18**.

In exemplary embodiments, the electronic display layer **14**, the illumination device **16**, and OL HX **20** may be provided within an access assembly **44** along with one or more components, such as, but not limited to, electronic circuits, air circulation devices **48**, sensors **54**, controller(s) **52**, power supplies, wiring, processors, video players, cameras, microphones, combinations thereof, and the like. The access assembly **44** may comprise a housing, framework, or one or more structural members, which may be attached to the housing **18** by way of one or more movement devices **46**, such as, but not limited to, hinges, springs, levers, pistons, combinations thereof, or the like configured to facilitate movement of the access assembly **44** between an opened position where the access assembly **44** is moved away from the housing **18** and a closed position where the access panel **44** is adjacent the housing **18**. One or more sealing devices **42**, such as, but not limited to, gaskets, may be provided between the housing **18** and the access assembly **44** to partially or completely seal when said access assembly **44** is placed in the closed position.

As illustrated with particular regard to at least FIG. 2C, the sealing devices **42**, such as but not limited to gaskets, may be provided between the cover panel **12** and the housing **18**. For example, without limitation, the sealing devices **42**, such as but not limited to gaskets, may extend at a first, preferably free end, to an interior surface of the cover panel **12** and may extend at a second, preferably fixed end, to a portion of the housing **18**. Stated another way, the cover panel **12** may extend above and below where the sealing device **42** meets the cover panel **12**. This exemplary arrangement, particularly placement of the sealing device **42** against the cover panel **12** instead of the housing **18**, may assist with reducing thermal transfer from external, ambient temperatures into the units **10**. In particular, because metal (of which the housing **18** may be comprised of in whole or in part) has a relatively high level of thermal conductivity where glass and/or polymer (of which the cover panel **12** may be comprised of in whole or in part) has a relatively low level of thermal conductivity. Thus, relatively cool temperatures of ambient air transferred to the housing **18** are less likely to be thermally conducted into relatively warm areas of the closed loop, by way of non-limiting example. This may reduce "cold spots" within the unit **10** where condensation is otherwise likely to form. The cover panel **12**, particularly when comprising glass and/or a polymer, may also provide a smoother surface for sealing, thereby providing a better seal, particularly compared to areas of metal where adhesive, welding, grinding, or the like may be provided.

In exemplary embodiments, such sealing devices **42** may be liquid impermeable, but may be vapor permeable. When moved into the open position, certain components of the electronic display assembly **10**, such as, but not limited to, a rear area of the OL HX **20**, the OL/CL HX **22**, customer

equipment, server racks, storage compartments, combinations thereof, or the like may be accessible. Furthermore, the access assembly **44** may be removed for service and/or replacement. Where more than one electronic display layer **14** is utilized, more than one access assembly **44** with the same or similar components may be provided and connected to the housing **18**.

Ambient air **56** may extend through one or more open loop airflow pathways within one or more of the OL HX **20**, the OL/CL HX **22**, combinations thereof, or the like. Such open loop airflow pathway(s) may be partially, mostly, substantially, or entirely sealed to separate ambient air **56** from circulating gas **58** traveling through one or more closed loop airflow pathways within one or more of the front air gap **13**, the OL/CL HX **22**, within the housing **18**, combinations thereof, or the like. In this way, particularly the ambient air **56** may be kept partially, mostly, substantially, or entirely separate from the circulating gas **58**.

One or more air circulation devices **48**, such as, but not limited to, fans, may be provided within or otherwise in fluid communication with the OL/CL HX **22**, the front air gap **13**, and/or other portions of one or more closed loop airflow pathways within the electronic display assembly **10** to cause the flow of circulating gas through some or all of the same when the air circulation devices **48** are operated. Such air circulation devices **48** may be in electronic communication with the same or different ones of the one or more controller(s) **52**. The air circulation devices **48** may comprise, for example, without limitation, axial fans, centrifugal fans, combinations thereof, or the like. Any number and/or type of air circulation devices **48** at any one or number of locations within the display assembly **10** may be utilized.

Alternatively, or additionally, the display assemblies **10** may comprise heaters, air conditioning units, filters, thermoelectric modules, heat sinks, combinations thereof, or the like.

The electronic display assembly **10** may comprise one or more sensors **54**. The sensors **54** may be provided at one or more locations at the electronic display assembly **10**. Some or all of the sensors **54** may be in electronic communication with one or more of the same or different controller(s) **52**. Such sensors **54** may include, for example, without limitation, location detection devices. Alternatively, or additionally, location data may be pre-programmed or updated manually. The sensor **54** may include temperature sensors, which may be located at intake(s), along the one or more open loop airflow pathways, at the exhaust(s), within the closed loop airflow pathways, at the illumination device **16**, at power supplies (which may be, for example, without limitation, located at and/or along rear surfaces of the illumination device **16**), at FPGA (Field Programmable Gate Array) die, at various framework or other components of the electronic display assembly **10**, at the one or more controller(s) **52**, a FPGA die, processor board, combinations thereof, or the like.

In exemplary embodiments, without limitation, at least one temperature and/or humidity sensor is located at, or proximate, an upper or uppermost surface of the housing **18** portion defining the closed loop airflow pathway, and at least one other temperature and/or humidity sensor is located at, or proximate, a lower or lowermost surface of the closed loop airflow pathway the housing **18** portion defining the closed loop airflow pathway. These may be, by way of non-limiting example, the coldest portions of the unit **10**, and thus the most likely to experience condensation. Using data from these locations may provide particularly conservative measurements for condensation control.

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Alternatively, or additionally, temperature and/or humidity sensors may be located at, or proximate, to an intake area, such as to detect the temperature of ingested ambient air.

These exemplary arrangements, and others, may assist with providing accurate information for determining internal unit 10 conditions, such as dewpoint spread for initiating modified operations. However, any number and location of temperature and/or humidity sensors may be utilized.

In exemplary embodiments, without limitation, multiple temperature readings may be taken at various portions of the assembly 10. Such values may be compared, such as but not limited to with respective offsets, and the lowest (coldest) values may be utilized by the controller(s) 52 to control the assembly 10 for purposes of operating the unit 10 normally or in the condensation mitigation mode. Offsets may be established to adjust for perceived inaccuracies, likely temperatures at other points in the units, localized temperature goals or needs, combinations thereof, or the like.

The sensors 54 may comprise one or more humidity sensors, which may be provided at the one or more open loop airflow pathways, the one or more closed loop pathways, one or more components of the display assembly 10, combinations thereof, or the like. The various sensors 54 may be configured to report readings data to the controller(s) 52. The size, shape, and/or location of the sensors 54 shown and/or described are merely exemplary and are not intended to be limiting. Any type, kind, and/or number of sensors 54 may be provided at any number of locations within the display assembly 10 to measure any number or type of data points. The humidity sensors, which may be configured to determine relative humidity, may include temperature sensors and absolute humidity sensors.

While illustrated internal to the display assembly 10, one or more of the sensors 54 and/or controller(s) 52 may be external to the display assembly 10. For example, without limitation, one or more of the sensors 54 may be located outside the housing 18. As another example, without limitation, some or all of the data points may be retrieved over one or more networks, such as the world wide web, from remote weather stations. The display assembly 10 may comprise one or more network communication devices 62 configured to retrieve such data, which may be periodically or continuously updated. As another example, without limitation, the controller(s) 52 may comprise one or more remote monitoring and/or control systems, such as, but not limited to, computers, smartphones, tablets, servers, combinations thereof, or the like, which may be in electronic communication with one or more controller(s) 52, processors, combinations thereof, or the like by way of one or more network connectivity devices. Data from both sensors 54 at the display assembly 10 and retrieved from outside sources may be utilized. For example, without limitation, data from outside sources may be retrieved by way of the network communication devices 62.

In exemplary embodiments, at least the controller 52 and/or network communication devices 62 are provided at the closed loop airflow pathways, though any location may be utilized.

FIG. 4 illustrates a flow chart with exemplary logic for operating the electronic display assembly 10 to control condensation. The relative humidity of air inside the display assembly 10 may be measured. For example, without limitation, the relative humidity may be determined by sampling data from the sensor(s) 54 configured to measure relative humidity at the display assembly 10. Alternatively, or additionally, the relative humidity may be determined by sampling data from the sensor(s) 54 configured to measure

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humidity at the display assembly 10 and the sensor(s) 54 may be configured to measure temperature at the display assembly 10, which may be the same or different from one another, and the relative humidity within the display assembly 10 may be calculated or determined from such data. Such calculation or determination may be made at the sensor(s) 54 or the controller(s) 52, such as based on a table or other data sources. Such temperature and/or humidity readings may be determined from sensor(s) 54 located at the one or more open loop airflow pathways of the ambient air, the one or more closed loop airflow pathways of the circulating gas, combinations thereof, or the like. Alternatively, or additionally, the temperature, humidity, and/or relative humidity may be retrieved from one or more network sources based on reported and/or measured conditions proximate a location of the electronic display assembly 10.

The dewpoint may be calculated from the relative humidity, humidity, and/or certain temperature data. The dewpoint may be calculated at the controller(s) 52 and/or sensor(s) 54, such as, for example, without limitation, by using formulas available at: <http://bmcnoldy.rsmas.miami.edu/Humidity.html> and/or <https://bmcnoldy.rsmas.miami.edu/mia/>. Any formula or algorithm for calculating dewpoint, known or yet to be developed, may be utilized. Alternatively, or additionally, the dewpoint may be retrieved from one or more network sources based on reported and/or measured conditions proximate the location of the electronic display assembly 10. The temperature used to calculate the dewpoint may be determined from sensor(s) 54 located at the one or more open loop airflow pathways of the ambient air, the one or more closed loop airflow pathways of the circulating gas, one or more network sources based on reported, measured conditions proximate a location of the electronic display assembly 10, combinations thereof, or the like.

The dewpoint spread may be calculated, such as by way of the controller(s) 52. The dewpoint spread may be calculated between the certain temperature data and the dewpoint. The temperature may be determined from sensor(s) 54 located at the one or more open loop airflow pathways of the ambient air, the one or more closed loop airflow pathways of the circulating gas, one or more network sources based on reported, measured conditions proximate a location of the electronic display assembly 10, combinations thereof, or the like. In exemplary embodiments, without limitation, the temperature may be determined as the lessor of sensor(s) 54 readings from within the electronic display assembly 10 or retrieved temperature conditions as retrieved from one or more network sources based on reported and/or measured conditions proximate the location of the electronic display assembly 10. Data from network sources shown and/or described herein may be retrieved, for example, without limitation, by way of the one or more network communication devices 62.

If the dewpoint spread is greater than or equal to $X^{\circ}\text{C.}$, which is variable but may be 2°C. , between 2°C. and 5°C. , 4°C. , or 5°C. for example, without limitation, a determination may be made, such as at the controller(s) 52, that no condensation is likely present. In such cases, the controller(s) 52 may be configured to operate the electronic display assembly 10 normally, such as under default operating parameters. If the dewpoint spread is greater than $Y^{\circ}\text{C.}$, which is variable but may be set of 0°C. , between 0°C. and 2°C. , or 2°C. in exemplary embodiments, and less than $X^{\circ}\text{C.}$, a determination may be made that condensation may or may not be present. In such cases, the controller(s) 52 may be configured to operate the electronic display assembly 10 in a first modified operating mode. If the dewpoint

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spread is less than or equal to $Y^{\circ}\text{C.}$, a determination may be made that condensation is likely present. In such cases, the controller(s) 52 may be configured to operate the electronic display assembly 10 in a second modified operating mode. The second modified operating mode may be the same or different from the first modified operating mode.

Other criteria, ranges, and/or thresholds may be utilized. For example, without limitation, other dewpoint spread criteria, ranges, and/or thresholds may be utilized. For example, without limitation, where a dewpoint spread equal to or less than $X^{\circ}\text{C.}$ is determined, the controller(s) 52 may be configured to command modified operations. As another example, without limitation, a half, quarter, or full degree margin of error may be utilized such that dewpoint spread equal to or less than $X+1^{\circ}\text{C.}$, by way of non-limiting example, may result in the controller(s) 52 commanding modified operations. These are just examples and are not intended to be limiting. Any number of thresholds and/or ranges and modified operating modes may be utilized.

The modified operating mode(s) (including, but not limited to, the first and second modified operating modes) may include commanding certain actions, such as, but not limited to, by way of the controller(s) 52, configured to raise the temperature of the electronic display assembly 10, thereby reducing the likelihood of condensation, and/or drive out moisture in the electronic display assembly 10. In exemplary embodiments, moisture may be driven out by increasing the temperature which may cause the air within the assembly, such as, but not limited to, the circulating gas 58 in the one or more closed loop airflow pathways, to expand. Because the closed loop airflow pathways are otherwise fully or partially sealed, this may result in driving out at least a portion of the circulating gas 58 from the one or more closed loop pathways, which may bring vaporized moisture with it. The air may permeate through one or more gaskets, which may be liquid tight but not necessarily vapor tight. In exemplary embodiments, the increased heat and/or airflow, such as from the one or more air circulation devices 48, may cause liquid moisture to vaporize or otherwise be gathered into the circulating gas which is then driven out of the display assembly 10.

The modified operating mode(s) may comprise commands to increase illumination at the illumination device 16, such as, but not limited to, driving the lighting elements 38 at an increased power level, reduce local dimming, and/or reduce dynamic dimming of the illumination device 16. This may be accomplished with or without altering operation of the electronic display layer 14. For example, without limitation, the electronic display layer 14 may be commanded to show a blank black screen despite increased illumination to increase heat without causing light pollution. As another example, without limitation, certain parameters of the electronic display layer 14, such as, but not limited to, grayscale may be altered to maintain essentially the same visible image characteristics while increasing the illumination. In these ways, the image displayed may appear unaltered to viewers, may conform to customer requirements, and/or prevent or reduce light pollution for example, without limitation. Alternatively, or additionally, limiting light leakage may maximize heat retention within the assembly 10.

The modified operating mode(s) may, alternatively, or additionally, comprise commands to reduce speed of and/or cease or otherwise minimize operation of air circulation devices 48 associated with the open loop airflow pathway(s). This may reduce the amount of relatively cool ambient air ingested into the assembly 10, which may cause temperatures to rise, or at least not lower as quickly. Operations of

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the air circulation devices 48 associated with the closed loop airflow pathway(s) may be modified as well. For example, without limitation, commands to increase the speed or and/or operation of such air circulation devices 48 may be increased, which may cause the circulating gas 58 to pick up condensation moisture.

In exemplary embodiments, once adequate temperatures, relative humidity, dewpoint, and/or dewpoint spread is reached, the modified operating mode(s) may be ceased and/or normal operations may be resumed. In exemplary embodiments, the commands shown and/or described herein may be carried out by the controller(s) 52.

Normal operating mode may be default mode or otherwise preprogrammed operating parameter, conditions, and/or logic. Such normal operating mode may permit relatively higher or unlimited air circulation device 48 speeds, run times, combinations thereof, or the like, particularly, but not limited to, for those air circulation devices 48 associated with the open loop airflow pathways.

Stated another way, where the dewpoint spread is less than X and/or Y (or another threshold), a determination may be made that the conditions are such that condensation is more likely to form. The modified operations, either of the first and/or second variety, may be automatically initiated by software routine which have the goal of increasing the dewpoint spread and/or otherwise mitigate the likelihood of condensation forming within the assemblies 10. Such modified operations may include adjusting the speed and/or thresholds at which air circulation devices 48 operate, adjusting illumination device 16 operations, combinations thereof, or the like. This may include increasing or decreasing fan speed. Such alterations may be made to acceptable fan speed ranges (e.g., 0-75% speed, 25-95% fan speed, etc.), fan speed limits (<25%, over 50 rpm, etc.), and/or operational speed setpoints (e.g., zero speed, 50% speed, 100% speed, etc.).

As illustrated with particular regard to at least FIG. 4B, alternatively or additionally, a single dewpoint spread threshold may be used such that the units 10 are configured to operate, such as by way of programmable control of the controller(s) 52, in either a normal operating mode or a condensation mitigation mode. For example, without limitation, where the dewpoint spread is determined, such as by the controller(s) 52, to be greater than “X” the controller(s) 52 may configure the unit 10 to operate normally, but where the dewpoint spread is determined to be less than “X” the controller(s) 52 may configure the unit 10 to operate in the condensation mitigation mode. When equal to “X”, the controller(s) 52 may configure the unit 10 to operate may be configured to either operate in the normal mode or the condensation mitigation mode. The controller(s) 52 may be configured to set the dewpoint spread to a threshold, such as between 1°C. and 10°C. The controller(s) 52 may configure cause the unit 10 to remain in the condensation mitigation mode unless and until the dewpoint spread increases above X by at least a predetermined margin, such as between 1°C. and 5°C. Alternatively, or additionally, the threshold for activating condensation mitigation mode may be different from deactivating condensation mitigation mode. For example, the threshold for transitioning from normal operating mode to condensation mitigation mode may be $X-A$ (e.g., 3°C.) while the threshold for transitioning from condensation mitigation mode to normal operating mode may be $X+B$ (e.g., 6°C.). An electronic notification may be generated and transmitted where the dewpoint spread reaches a predetermined threshold below “X” (e.g., $X-C$,

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such as 0° C.). This may reduce or prevent rapid transitions between condensation mitigation mode and normal operating mode.

FIG. 5 illustrates a flow chart with other exemplary logic for operating the electronic display assembly 10 to minimize, reduce, control and/or eliminate condensation. The steps shown and/or described with respect to FIG. 5 may be the same or similar to those shown and/or described with respect to FIG. 4 except as otherwise specified herein. For example, without limitation, if a determination is made that condensation may or may not be present or that condensation is likely present, a determination may be made, such as at the controller(s) 52, whether certain safety thresholds are met and/or exceeded. If such safety thresholds are met and/or exceeded, the controller(s) 52 may command the electronic display assembly 10 to operate normally (i.e., without modified operations configured to minimize, reduce, control, and/or eliminate condensation). If some or all of the safety thresholds are not met and/or exceeded, the controller(s) 52 may command the electronic display assembly 10 to initiate modified operations, which may be the same, or different from the modified operations shown and/or described with respect to FIG. 4.

FIG. 6 illustrates an exemplary interface 64. The interface 64 may be provided at the controller(s) 52 and/or one or more remote electronic devices. The interface 64 may be configured to accept certain user input regarding safety thresholds, modified operations, normal operations, and other operations of the assembly 10. For example, without limitation, such criteria may be selected at one or more remote electronic devices, such as, but not limited to, personal computers, server computers, smartphones, tablets, combinations thereof, or the like and transmitted to the assemblies 10 by way of the network communication devices 62.

In exemplary embodiments, the controller(s) 52 may be configured to first temporarily initiate normal operations upon determination that the safety thresholds are met and/or exceeded. This may cause air circulation devices 48, such as those associated with the open loop airflow pathways, to partially or fully operate at higher levels (e.g., speed, run-times, etc.), ingesting relatively more ambient air in attempt to cool the assembly 10 for a period of time. Normal operations may be resumed for a period of time, such as, but not limited to, 120 seconds, though the amount of time may be variable and may be programmed and/or altered, such as at value 63. If after the period of time safety thresholds are no longer met and/or exceeded, modified operations may be resumed. If the safety thresholds are still met and/or exceeded after the period of time, then the assemblies 10 may be configured to continue normal operations for at least a period of time, which may be fixed or indefinite. The period of time may be programmed at the time parameter of the interface 64, for example, without limitation. The safety thresholds may be set, for example, without limitation, at the one or more safety parameters 68 of the interface 64. In exemplary embodiments, only a single temporary initiation of normal operations may be permitted before a longer term or fixed return to normal operations is commanded, such as by the controller(s) 52.

The source for dewpoint calculations may be set at a dewpoint source parameter 70 of the interface 64. For example, without limitation, the source may be selected between an internal relative humidity sensor, remote sources, combinations thereof, or the like.

Any of the variables, parameters, conditions, combinations thereof, or the like may be pre-programmed and/or

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programmed at the same or different interfaces 64, such as, but not limited to, at the controller(s) 52 and/or by remote devices and the network communication devices 62.

FIG. 6B illustrates another exemplary embodiment of the interface 64. The interface may include various offset values 71. Offset values 71 may be values added to measured or determined values, such as at the controller(s) 52, for comparison against a respective threshold. These may be set to reflect errors in measurement, differences from perceived values elsewhere, combinations thereof, or the like. A delay value 73 may be set which delays, such as by way of the controller(s) 52, implementation of condensation mitigation mode, such as for a specified number of seconds, minutes, combinations thereof, or the like. This may allow the assembly 10 some time to adjust to relatively minor variations without changing operating modes, eliminate erroneous readings, combinations thereof, or the like. A hysteresis value 75 may be established which acts as a margin which the dewpoint spread must clear before the controller(s) 52 switch operating modes. This may prevent rapid or frequent changing between operating modes. A dewpoint spread minimum value 75 may be abolished. This may be dewpoint spread value at which the controller(s) 52 is configured to automatically generate and/or transmit an electronic notification, which may indicate conditions likely to result in condensation. All of the values shown and/or described herein may be set by software program, may be variable, and/or may be locally and/or remotely updated from time to time, such as by receipt of one or more authenticated commands.

As further illustrated in FIG. 7A, in exemplary embodiments, the controller(s) 52 may be configured to initiate modified or normal operations based upon a dewpoint spread threshold parameter 74 and/or a buffer 72, one or both of which may be programmable at the interface 64. The dewpoint spread threshold parameter 74 and/or buffer 72 may be utilized to determine a dewpoint spread, such as is shown and/or described herein. The controller(s) 52 may be configured to command the assembly 10 to utilize modified operations where the determined dewpoint spread meets the dewpoint spread threshold parameter 74 and/or exceeds the dewpoint spread threshold parameter 74 by at least, or more than, the buffer 72.

The controller(s) 52 may be configured to command the assembly 10 to continue utilizing modified operations until such a time as the determined dewpoint spread falls below the dewpoint spread threshold parameter 74, such as by more than the buffer 72. The controller(s) 52 may be configured to command the assembly 10 to utilize normal operations where the determined dewpoint spread does not meet the dewpoint spread threshold parameter 74 and/or is below the dewpoint spread threshold parameter 74 by at least the buffer 72. The controller(s) 52 may be configured to command the assembly 10 to continue utilizing normal operations until such as time as the determined dewpoint spread exceeds the dewpoint spread threshold parameter 74 by at least the buffer 72. In this way, a programmable buffer may be provided in either direction of temperature change against the programmable threshold. However, the safety thresholds may still be considered and prioritized such that the assembly 10 defaults to normal operations where one or more of the safety thresholds are met and/or exceeded.

Alternatively, or additionally, the dewpoint spread set buffer 72 may comprise the dewpoint spread value at which the controller(s) 52 will cause the assembly 10 to move into condensation mitigation mode. Stated another way, the dewpoint spread set buffer 72 may be the value J, which

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when dropped below may initiate condensation control measures. The dewpoint spread clear buffer 74 may comprise the dewpoint spread value at which the controller(s) 52 will cause the assembly 10 to return to normal operating mode. Stated another way, the dewpoint spread clear buffer 74 may be the value K, which when raised above may return the unit 10 to normal operations. In this fashion, the value which activates and deactivates condensation mitigation mode may be different.

As further illustrated in FIG. 7B, in exemplary embodiments, an attempt to bring the assembly 10 back below or otherwise within the safety thresholds may be provided, such as by way of the programmable time parameter. The controller(s) 52 may be configured to temporarily operate the assembly 10 in normal operations for the time parameter. If, after the time parameter is reached, and one or more of the safety thresholds are still exceeded or readings are otherwise outside the safety thresholds, then the controller(s) 52 may move the assembly 10 to normal operations until such as time as the threshold parameters 74 are exceeded by at least the buffer 72 and readings are within the safety thresholds. If, after the time parameter is reached, readings are within the safety thresholds, then the controller(s) 52 may resume modified operations until such a time as the threshold parameters 74 are exceeded by at least the buffer 72 or readings are outside the safety thresholds. This may give the assembly 10 a chance to recover from slipped operations and/or essentially disregard a potentially erroneous and/or outlier reading. This attempt may be only completed one time before commanding normal operations. For example, only a single loop of the command logic may be permitted.

The safety thresholds may include, for example, without limitation, temperatures above or below certain thresholds, within certain ranges, combinations thereof, or the like, such as measured by at least certain of the sensors 54. In exemplary embodiments, without limitation, the safety thresholds may include some or all of the following: temperature as measured by one or more sensors 54 within or adjacent to the closed loop airflow pathways being below 30° C., temperature as measured by one or more sensors 54 at or in proximity to the illumination device 16 below 40° C., PS (Power Supply) Max is below 50° C., FPGA is below 70° C., controller(s) 52 is/are below 50° C., sensor(s) 52 measuring humidity is/are below 35° C., combinations thereof, or the like. Other criteria and/or thresholds may be utilized, such as, but not limited to, other temperatures from other locations and/or at different thresholds. The safety thresholds, for example, without limitation, may be used to more accurately determine internal temperature of the display assembly 10 and thus may provide a more accurate determination of the likelihood that condensation is present and thus whether modified operations should be undertaken. The safety thresholds in exemplary embodiments may be variable and programmable, such as, but not limited to, at the one or more safety parameters 68 of the interface 64. The safety parameters 68, dewpoint source parameter 70, dewpoint spread threshold parameter 74, buffer 72, and/or time parameter shown in FIG. 6 represent exemplary operating parameters and are provided for example, without limitation.

In exemplary embodiments, modified operations may be provided regularly, such as during transitions from nighttime to daylight hours, following the end of power efficiency modes, during cold temperatures, combinations thereof, or the like. In other exemplary embodiments, modified operations may be dependent on ambient conditions and/or readings from the sensors 54. For example, without limitation,

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modified operating operations configured to increase the internal temperature of the electronic display assembly 10, such as, but not limited to, increasing power to illumination device 16, reducing speed of air circulation devices 48, during or following relatively cool nighttime hours and/or days or times with relatively cool ambient temperatures, and modified operations configured to drive out moisture, such as, but not limited to, increased speed of air circulation devices 48, during relatively warm daytime hours and/or days or times with relatively warm ambient temperatures.

Modified operations may include changing operational speed of air circulation devices 48 of the unit 10. In exemplary embodiments, without limitation, modified operations may include decreasing operational speed of air circulation devices 48 associated with open loop airflow pathways and/or ambient air, such as down to 20% (of maximum normal operation speed) or zero, though any speed, speed reduction, speed range, or the like may be utilized. In this fashion, flow of typically relatively cooler ambient air may be decreased, which tends to warm the units 10. Alternatively, or additionally, modified operations may include increasing operational speed of air circulation devices 48 associated with closed loop airflow pathways or areas and/or circulating gas, such as up to 80% or 100% (of maximum normal operation speed), though any speed, speed reduction, speed range, or the like may be utilized. This may assist with driving moisture out of closed areas, such as any closed loop airflow pathways, sealed, or semi-sealed areas, homogenizing internal air temperatures, and/or promoting efficient thermal transfer, by way of non-limiting example.

In other exemplary embodiments, the temperature, humidity, relative humidity, dew point spread, or other data may be derived from predicted weather data, such as based on historical patterns, from internet-based sources, combinations thereof, or the like. In such cases, certain modified operating modes may be scheduled and/or initiated as preventative measures based on predicted data.

The various measures shown and/or described herein, including, but not limited to, humidity measures, temperature measures, combinations thereof, or the like, may be determined by way of multiple measurements from the same or different sensors 54, internet-based sources or other remote measures or user input, combinations thereof, or the like, and the utilized measures may be an average, highest, lowest, mean, median value, combinations thereof, or the like.

In exemplary embodiments, some or all air circulation devices 48 may be kept at a minimum speed, such as regardless of normal or modified operations, to provide relatively uniform temperature within the electronic display assembly 10, consistent readings at sensors 52, combinations thereof, or the like. In exemplary embodiments, without limitation, this minimum speed may be 12% of maximum possible normal operating speed.

As illustrated with particular regard to at least FIG. 7C, the safety parameters analysis is optional. This applies to all embodiments shown and/or described herein.

FIG. 8 provides other exemplary logic for operating the display assembly 10. It may be desirable to operate the air circulating devices 48 within the display assembly 10 to control temperatures within the display assembly 10 and/or provide condensation control. In exemplary embodiments, each of the air circulating devices 48 may comprise one or more fans, such as provided in banks or sets. Each of the air circulating devices 48 may be associated with one or more of the sensors 54. Each of the sensors 54 may be configured to measure temperature of proximate air.

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Zones may be virtually defined within the display assemblies 10, such as at the controller(s) 52. Each zone may be associated with one or more of the air circulating devices 48 and/or one or more of the sensors 54. For example, without limitation, one zone may comprise front air gap 13, another zone may include the OL HX 20, another zone may include the OL/CL HX 22, combinations thereof, or the like. Within the front air gap 13, a first zone may be defined between the electronic display layer 14 and the cover panel 12, which may also be referred to as the LCD cavity, and a second zone may be defined between the electronic display layer 14 and the illumination device 16, which may also be referred to as the LED cavity. Any number of zones may be defined within the display assemblies 10.

Operational ranges for the air circulating devices 48 may be established, such as at the controller(s) 52. Each of the operational ranges may be associated with one or more of the air circulating devices 48. Desired operating ranges may be established for the sensors 54, such as at the controller(s) 52. Each of the desired operating ranges may be associated with one or more of the sensors 54. Operational ranges for the air circulating devices 48 and/or desired operating ranges for the sensors 54 may be specific to the date, time, ambient conditions, combinations thereof, or the like, and may be programmed at, or stored at, the controller(s) 52. Operational ranges for the air circulation devices 48 and/or desired operating ranges for the sensors may be specific to the zone, air circulation device 48, and/or sensor 54 or for the whole display assembly 10. For example, without limitation, operational ranges for the air circulating devices 48 and/or desired operating ranges for the sensors 54 may be specific to day time or night time operations. Such day time and/or night time operations may be determined based on a location of the display assembly 10 and/or time of year (e.g., reflecting sunrise and/or sunset times based on location and date). In exemplary embodiments, without limitation, operational ranges for the air circulating devices 48 associated with the closed loop airflow pathway(s) and/or circulating gas 58 may be set to 100% fan speed during daytime hours and 15-100% during nighttime hours, and operational ranges for the air circulating devices 48 associated with the open loop airflow pathway(s) and/or ambient air 56 may be 0-100% at all times.

Readings from the sensors 54 may be taken, such as continuously, periodically, sporadically, combinations thereof, or the like. Operation (e.g., speed, number of active fans, volumetric flow rate, power supplied, etc.) of some or all of the air circulation devices 48 may be controlled by a highest of readings from the sensors 54 at a given time or period of time relative to the associated desired operating range for the various sensors 54. For example, without limitation, the zone and/or sensor 54 having a highest reading relative to the maximum limit of the desired operating range associated with each zone or sensor 54 may be used by the controller(s) 52 to set the operating conditions of the air circulating devices 48 with the associated operational range. Such sensor 54 and/or zone may be controlling until subsequent readings indicate return to the desired operating range. Alternatively, or additionally, such sensor 54 and/or zone may be controlling until another sensor 54 and/or zone becomes highest and/or the furthest outside of the associated desired operating range. This may enhance thermal management by ensuring that the most problematic zone or sensor 54 reading is driving operations.

The controller(s) 52 may be configured to ramp speed of the air circulating device(s) 48 up or down on a linear basis, inversely proportional ratio, by some multiple or other ratio

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relative to how far the temperature is from the desired operating range, combinations thereof, or the like. Such adjustments may be made incrementally and readings retaken and adjustments made accordingly.

Such control may be performed on a zone-by-zone basis or for the entire display assembly 10. For example, without limitation, air circulating devices 48 within a given zone and/or associate sensor(s) 54 may be adjusted individually based on such readings, or the air circulation devices 48 for the entire assembly 10 may be adjusted based on such readings, even if from a single zone and/or sensor 54.

Where a maximum operating temperature is reached or exceeded at one or more of the sensors 54, operations of the air circulation devices 48 may be adjusted up to a maximum operational level (e.g., speed, number of active fans, volumetric flow rate, power supplied, etc.) within the operational ranges. If readings from the sensors 52 indicate that the maximum operating temperature subsequently remains reached or exceeded, power to the illumination device 16, such as a backlight, may be reduced until temperatures fall below the maximum operating temperatures. Such reduction may be made in an inversely proportional fashion to how far the temperature has exceeded the maximum operating temperature. Such reduction may be made incrementally and adjustments made accordingly.

Sensor readings may be continuously or periodically retaken and operations adjusted accordingly.

Temperature readings from the sensors 54 may be communicated to the controller(s) 52 which may be configured to make operational determinations and adjustments for said air circulation devices 48 based on said readings. Alternatively, or additionally, such readings may be transmitted, such as by the network communication devices 62, to one or more remote controller(s) 52 located remote from the display assembly 10. Updates to the operational ranges for the air circulating devices 48 and/or desired operating ranges for the sensors 54 may be made from time to time, such as by way of instructions communicated to the controller(s) 52 through the network communication device 62 from one or more remote devices.

The control logic shown and/or described with respect to the several figures and accompanying description provided herein may be used together or separately. For example, without limitation, the control logic shown and/or described with regard to FIG. 8 may be used with the condensation control logic of FIGS. 4-7B or separately therefrom.

While certain measures are shown and/or described herein in terms of degrees Celsius, equivalent measures in degrees Fahrenheit, Kelvin, or other measurement standards may be utilized.

The ambient air 56 within the open loop airflow pathway(s) may be entirely or substantially prevented from mixing with the circulating gas 58 of the closed loop airflow pathway(s). For example, without limitation, the display assembly 10 may be configured to comply with various ingress protection standards, such as, but not limited to, IP 65, IP 66, IP 67, IP 68, combinations thereof, or the like, at least with regard to the closed loop airflow pathway(s) or other particular areas of the assembly 10. Ambient air 56 may comprise air ingested from the surrounding environment and may or may not be filtered. The circulating gas 58 may comprise air kept fully or partially separate from the ambient air 56 in exemplary embodiments. For example, the circulating gas 58 may include ambient air 56 trapped when the assembly 10 is formed or otherwise periodically accessed (e.g., for servicing). Alternatively, or additionally, the circulating gas 58 may comprise filtered or purified air.

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Modified operations and/or condensation mitigation mode, which may be one and the same, may include establishing different temperature levels and/or offsets at which the fans or other thermal management components are activated and/or are operated. For example, without limitation, the fans or other thermal management components may be configured to operate relative to various temperature readings. The activation/deactivation points, operational speed and/or levels, combinations thereof, or the like of such fans or other thermal management components may be varied between normal operating mode and modified operations and/or condensation mitigation mode. In exemplary embodiment, without limitation, fan speed or other operational characteristics of fans or other thermal management components may be set relative to temperature (e.g., at X temperature, Y fan speed, at Z temperature, A fan speed, etc.). Such relationship may be established by table, linear relationship, exponentiation relationship, logarithmic relationship, algorithmic relationship, combinations thereof, or the like. Certain preferred operating parameters, thresholds, or the like are disclosed herein which result in advantageous operation but are not necessarily intended to be limiting.

Any embodiment of the present invention may include any of the features of the other embodiments of the present invention. The exemplary embodiments herein disclosed are not intended to be exhaustive or to unnecessarily limit the scope of the invention. The exemplary embodiments were chosen and described in order to explain the principles of the present invention so that others skilled in the art may practice the invention. Having shown and described exemplary embodiments of the present invention, those skilled in the art will realize that many variations and modifications may be made to the described invention. Many of those variations and modifications will provide the same result and fall within the spirit of the claimed invention. It is the intention, therefore, to limit the invention only as indicated by the scope of the claims.

Certain operations described herein, may be performed by one or more electronic devices. Each electronic device may comprise one or more processors, electronic storage devices, executable software instructions, and the like, configured to perform the operations described herein. The electronic devices may be general purpose computers or specialized computing devices. The electronic devices may comprise personal computers, smartphones, tablets, databases, servers, or the like. The electronic connections and transmissions described herein may be accomplished by wired or wireless means. The computerized hardware, software, components, systems, steps, methods, and/or processes described herein may serve to improve the speed of the computerized hardware, software, systems, steps, methods, and/or processes described herein.

What is claimed is:

1. A display assembly providing condensation mitigation, said display assembly comprising:

a housing;

a cover layer forming a forward portion of the housing, wherein said cover layer is transparent or translucent; an electronic display located within the housing rearward of the cover layer, wherein said cover layer has an outer perimeter which extends beyond an outer perimeter of the electronic display;

a continuous airflow pathway extending entirely within the housing such that the continuous airflow pathway is partitioned from an ambient environment, wherein the

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continuous airflow pathway extends, at least in part, forward and rearward of the electronic display to form a circuit; and

one or more gaskets provided to, at least in part, partition said continuous airflow pathway from the ambient environment, wherein said one or more gaskets extend from an interior surface of the cover layer to a portion of the housing interior to said cover layer without contacting said electronic display so as to provide one or more gaps fluidly connecting portions of the continuous airflow pathway extending forward and rearward of the electronic display as part of said circuit.

2. The display assembly of claim 1 further comprising: one or more sensors located internal to said housing; and a controller in electronic communication with said one or more sensors and comprising software instructions stored at one or more non-transitory electronic storage devices, which when executed, configure one or more processors to:

determine a dewpoint temperature for air at the display assembly;

determine an internal temperature for the display assembly based, at least in part, on data received from said one or more sensors;

calculate a dewpoint spread between said dewpoint temperature and said internal temperature;

where said dewpoint spread is equal to or below a first predetermined value, initiate modified operations; and

where said dewpoint spread is equal to or above a second predetermined value, cease said modified operations.

3. The display assembly of claim 2 further comprising: an additional airflow pathway extending through at least a portion of said housing; and

one or more fans located along said additional airflow pathway, wherein said modified operations comprise changing an operating speed for said one or more fans.

4. The display assembly of claim 3 wherein:

said modified operations comprise reducing the operating speed for said one or more fans in accordance with a predetermined algorithm.

5. The display assembly of claim 3 further comprising: an intake and an exhaust located at said housing and fluidly connected to an ambient environment, wherein said additional airflow pathway comprises an open loop airflow pathway extending between said intake and said exhaust for ambient air; and

one or more additional fans located along the additional airflow pathway, wherein said modified operations comprise modifying an operating speed for said one or more additional fans.

6. The display assembly of claim 5 wherein:

said modified operations comprise reducing the operating speed for said one or more fans in accordance with a predetermined algorithm and increasing the operating speed for said one or more additional fans in accordance with the predetermined algorithm.

7. The display assembly of claim 2 wherein:

said electronic display comprises one or more illumination devices; and

said modified operations comprise increasing power supplied to the one or more illumination devices.

8. The display assembly of claim 2 further comprising: an additional airflow pathway extending through at least a portion of said housing; and

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said one or more sensors comprises a first sensor located within the additional airflow pathway.

9. The display assembly of claim 8 wherein:

said additional airflow pathway is fluidly connected to an ambient environment;

said one or more sensors comprises a second sensor located within the continuous airflow pathway;

the data received from said second sensor is utilized by said controller to determine the internal temperature for the display assembly; and

the data received from the first sensor is utilized by said controller to determine the dewpoint temperature.

10. The display assembly of claim 2 wherein:

said one or more sensors comprise multiple temperature sensors; and

the controller is configured to utilize data from a lowest reporting one of said multiple temperature sensors to determine the internal temperature.

11. The display assembly of claim 2 wherein:

at least one of the one or more sensors comprises a humidity sensor; and

the controller is configured to receive data from said humidity sensor to determine the dewpoint temperature.

12. The display assembly of claim 2 wherein:

the controller is configured to utilize data from an internet-based weather reporting service to determine the dewpoint temperature.

13. The display assembly of claim 2 wherein:

the controller is configured to receive data from one or more remote electronic device specifying or revising the first predetermined value and the second predetermined value.

14. The display assembly of claim 1 wherein:

said electronic display comprises a liquid crystal display and light emitting diodes providing direct backlighting to the liquid crystal display; and

the continuous airflow pathway is partitioned from the ambient environment in accordance with at least one of: ingress protection code 65, 66, 67, and 68.

15. The display assembly of claim 1 further comprising: additional software instructions stored at said one or more non-transitory electronic storage devices, which when executed, configure said one or more processors to: if a predetermined maximum internal temperature threshold is met or exceeded, reduce a power supplied to a backlight for the electronic display.

16. The display assembly of claim 1 wherein:

each of the one or more gaskets comprise a rubberized material; and

each of the one or more gaskets extend exclusively between the interior surface of the cover layer and the portion of the housing interior to said cover layer.

17. A display assembly providing condensation mitigation, said display assembly comprising:

a housing comprising an intake and an exhaust;

a cover layer forming a forward portion of the housing; an electronic display within the housing rearward of the cover layer;

a first airflow pathway for ambient air extending within the housing between the intake and the exhaust;

a second, continuous airflow pathway extending entirely within the housing such that the second airflow pathway is partitioned from an ambient environment;

one or more sensors located internal to said housing; and one or more gaskets provided to, at least in part, partition said continuous airflow pathway from said first airflow

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pathway and the ambient environment, wherein said one or more gaskets extend from an interior surface of the cover layer to a portion of the housing.

18. The display assembly of claim 17 further comprising:

a controller in electronic communication with said one or more sensors and comprising software instructions stored at one or more non-transitory electronic storage devices, which when executed, configure one or more processors to:

determine a dewpoint temperature for air at the display assembly;

determine an internal temperature for the display assembly based, at least in part, on data received from said one or more sensors;

calculate a dewpoint spread between said dewpoint temperature and said internal temperature;

where said dewpoint spread is equal to or below a first predetermined value, initiate modified operations; and

where said dewpoint spread is equal to or above a second predetermined value, cease said modified operations.

19. A display assembly providing condensation mitigation, said display assembly comprising:

a housing comprising a chassis or framework comprising metal;

a cover layer forming a forward portion of the housing, wherein said cover layer is transparent or translucent and comprises at least one of: glass and a polymer;

an electronic display for displaying electronic images located within the housing rearward of the cover layer, wherein said cover layer has an outer perimeter which extends beyond an outer perimeter of the electronic display such that the electronic images generated at the electronic display are visible through the cover layer;

a continuous airflow pathway partitioned from an ambient environment and extending within the housing, including in front of the electronic display and behind the electronic display to form a circuit;

at least one air circulation device within the continuous airflow pathway for moving gas through the continuous airflow pathway when activated; and

one or more gaskets provided to, at least in part, partition said continuous airflow pathway from the ambient environment, wherein each of said one or more gaskets extend from an interior surface of a portion of the cover layer located beyond an outer perimeter of the electronic display to a portion of the chassis or framework, without contacting said electronic display, so as to provide gaps for the gas of the continuous airflow pathway along edges of the electronic display, thereby fluidly connecting portions of the continuous airflow pathway extending in front of, and behind, the electronic display as part of said circuit;

wherein said continuous airflow pathway is partitioned from the ambient environment in accordance with at least one of: ingress protection code 65, 66, 67, and 68, at least in part by said one or more gaskets.

20. The display assembly of claim 19 wherein:

each of the one or more gaskets comprise a rubberized material; and

each of the one or more gaskets extends exclusively between the interior surface of the portion of the cover layer and the portion of the chassis or framework.